

Milestone_5_Exploratory_Data_Analysis_and_Visualization

May 29, 2026

1 Milestone 5: Exploratory Data Analysis and Visualization

Merging the Data and Storing in a Database/Visualizing Data

Now that you have cleaned and transformed your 3 datasets, you need to load them into a database. You can choose what kind of database (SQLite or MySQL, Postgre SQL are all free options). You will want to load each dataset into SQL Lite as an individual table and then you must join the datasets together with SQL joins into 1 dataset.

Once all the data is merged together in your database, create 5 visualizations that demonstrate the data you have cleansed. You should have at least 2 visualizations that have data from more than one source, using SQL joins (meaning, if you have 3 tables, you must have visualizations that span across 2 of the tables – you are also welcome to use your consolidated dataset that you created in the previous step, if you do that, you have met this requirement).

For the visualization portion of the project, you are welcome to use a python library like Matplotlib, Seaborn, or an R package ggPlot2, Plotly, or Tableau/PowerBI.

PowerBI is a free tool that could be used – Tableau only has a free web author. If you use Tableau/PowerBI you need to submit a PDF with your assignment vs the Tableau/PowerBI file.

Clearly label each visualization. Submit your code for merging and storing in the database, with your code for the visualizations along with a 250-500-word summary of what you learned and had to do to complete the project. In your write-up, make sure to address the ethical implications of cleansing data and your project topic. You can submit a Jupyter Notebook or a PDF of your code. If you submit a .py file you need to also include a PDF or attachment of your results.

You must submit the following: - Jupyter Notebook File or PDF of your code with Milestone # listed. - Each step and visualization should be clearly labeled. - Human readable dataset after all transformations should be printed at the end of your notebook. - 250-500 word summary of what you learned and a summary of the ethical implications answering the following questions: - What changes were made to the data? - Are there any legal or regulatory guidelines for your data or project topic? - What risks could be created based on the transformations done? - Did you make any assumptions in cleaning/transforming the data? - How was your data sourced / verified for credibility? - Was your data acquired in an ethical way? - How would you mitigate any of the ethical implications you have identified?

```
[1]: # Import the necessary libraries.  
import pandas as pd  
import sqlite3  
import matplotlib.pyplot as plt
```

```
import seaborn as sns
from textblob import TextBlob
from wordcloud import WordCloud

# import numpy as np
```

```
[2]: # Define the file paths for IMDb, cleaned_movies and Rotten Tomatoes clean
      ↪ datasets.
file1 = 'cleaned_imdb_df.csv'
file2 = 'cleaned_movie_dataset.csv'
file3 = 'rt_cleaned_movies_df.csv'

# Read CSV files (IMDb, cleaned_movies, and Rotten Tomatoes) into a pandas
      ↪ DataFrame.
df1 = pd.read_csv(file1)
df2 = pd.read_csv(file2)
df3 = pd.read_csv(file3)

# Connect or create to an SQLite database if it doesn't exist.
conn = sqlite3.connect('movies_database.db')

# Write each DataFrame to the database.
df1.to_sql('imdb_data', conn, if_exists='replace', index=False)
df2.to_sql('movie_data', conn, if_exists='replace', index=False)
df3.to_sql('rt_data', conn, if_exists='replace', index=False)

# Verify the tables were created.
cursor = conn.cursor()
cursor.execute("SELECT name FROM sqlite_master WHERE type='table';")
tables = cursor.fetchall()
print("Tables in database:", tables)

# Commit and close connection.
conn.commit()
```

Tables in database: [('imdb_data',), ('movie_data',), ('rt_data',)]

```
[3]: # SQL query to join tables based on corrected column names.
query = """
SELECT
    imdb.*,
    movie_data.*,
    rt_movies.*
FROM
    imdb_data AS imdb
LEFT JOIN
    movie_data AS movie_data
```

```

ON
    imdb.title = movie_data.title
LEFT JOIN
    rt_data AS rt_movies
ON
    imdb.title = rt_movies.[Movie Title];
"""

# Execute the join query.
joined_df = pd.read_sql_query(query, conn)

# Save the result as a new CSV file.
output_path = 'joined_movies_dataset.csv'
joined_df.to_csv(output_path, index=False)

# Close the connection.
conn.close()

output_path

```

[3]: 'joined_movies_dataset.csv'

[4]: # Read the 'joined_movies_dataset.csv' file into a pandas DataFrame.
joined_movies_dataset = pd.read_csv('joined_movies_dataset.csv')

[5]: # Group the data by 'genre', sum the 'audience_count' for each genre,
get the top 10 genres with the highest audience counts, and reset the index.

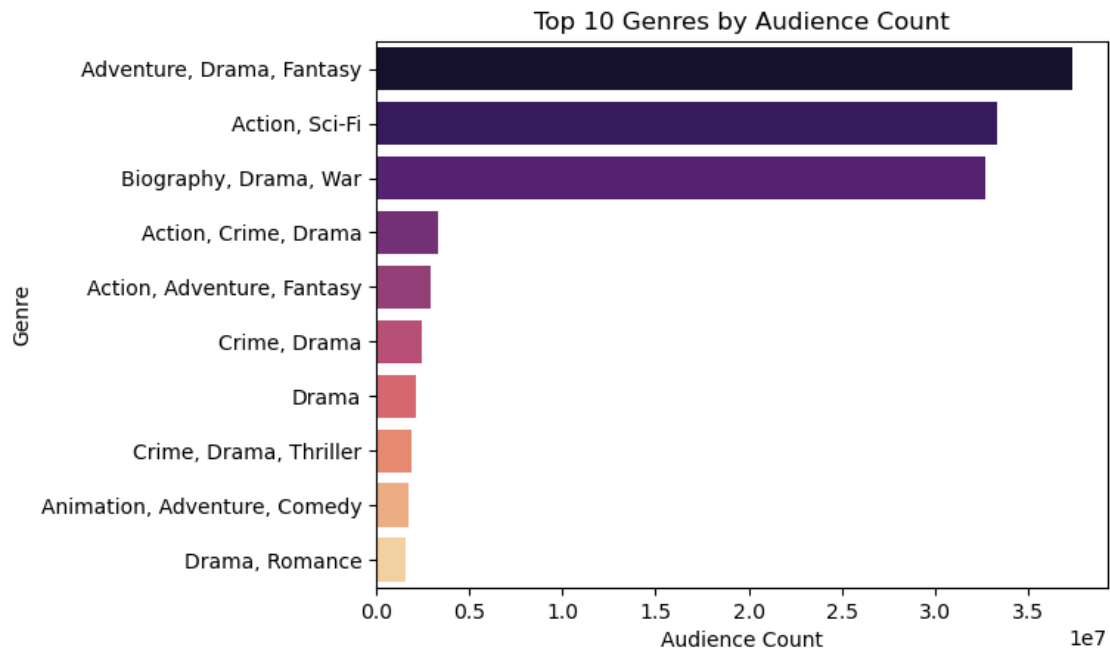
```

genre_audience = (
    joined_movies_dataset.groupby('genre')['Audience Count']
        .sum()
        .nlargest(10)
        .reset_index()
)

# Plotting the top 10 genres by audience count using seaborn's barplot.
sns.barplot(data=genre_audience, x='Audience Count', y='genre', hue='genre',
            palette='magma', legend=False)

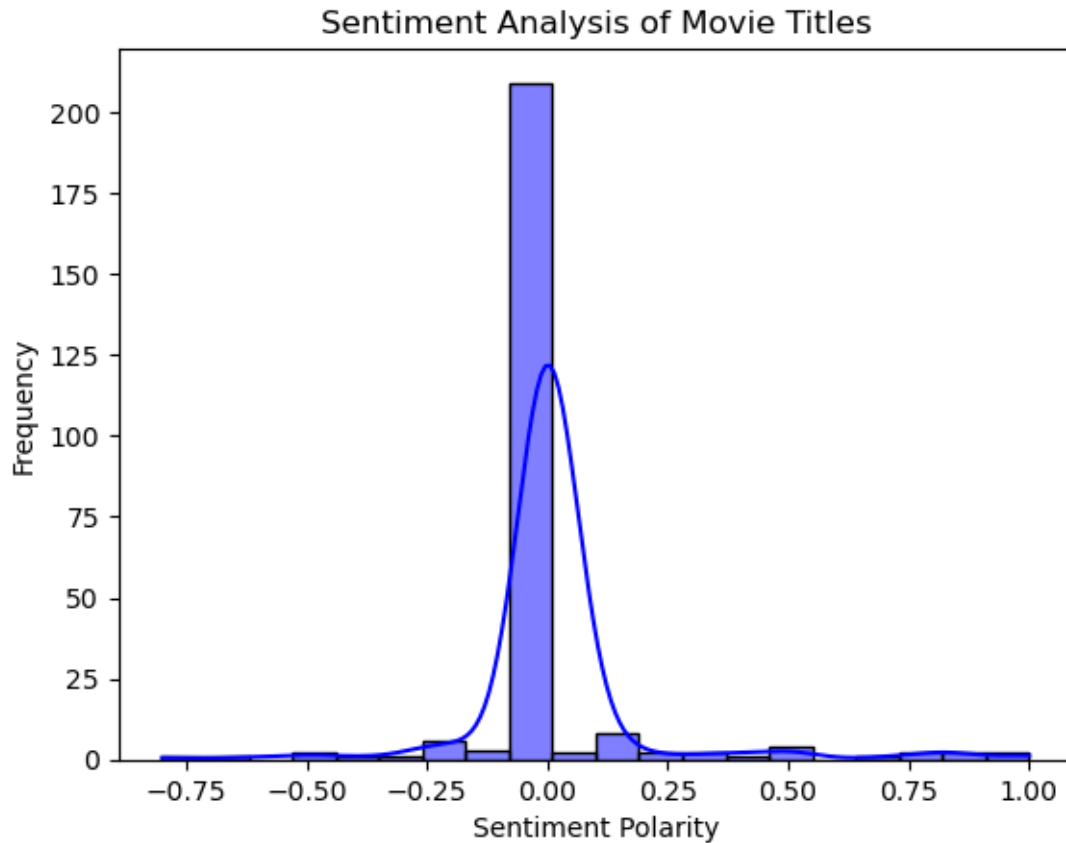
plt.title("Top 10 Genres by Audience Count")
plt.xlabel("Audience Count")
plt.ylabel("Genre")
plt.show()

```



```
[6]: # Perform sentiment analysis to each movie title using TextBlob.
# 'TextBlob(x).sentiment.polarity' calculates the sentiment polarity of the
    ↪ title.
# The result is stored in a new column called 'title_sentiment'
joined_movies_dataset['title_sentiment'] = joined_movies_dataset['title'].
    ↪ apply(lambda x: TextBlob(x).sentiment.polarity)

# Plot sentiment distribution.
# The 'histplot' function creates a histogram with a Kernel Density Estimate
    ↪ (KDE) overlay
sns.histplot(joined_movies_dataset['title_sentiment'], kde=True, bins=20,
    ↪ color='blue')
plt.title("Sentiment Analysis of Movie Titles")
plt.xlabel("Sentiment Polarity")
plt.ylabel("Frequency")
plt.show()
```

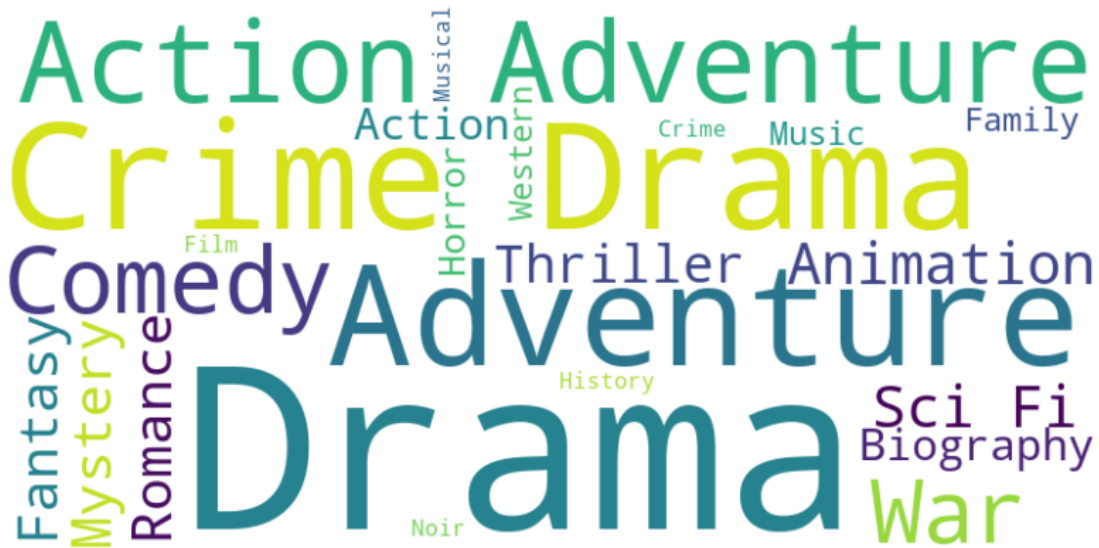


```
[7]: # Combine all the genres into a single string, removing any missing values
      ↳ (NaNs).
      # The 'join' method merges all genre entries from the 'genre' column into one
      ↳ long string
      all_genres = ' '.join(joined_movies_dataset['genre'].dropna())

      # Generate a word cloud from the combined genre string.
      # The WordCloud object takes the text (all_genres) and creates a visual
      ↳ representation.
      wordcloud = WordCloud(width=800, height=400, background_color='white').
        ↳ generate(all_genres)

      # Plot
      plt.figure(figsize=(10, 6))
      plt.imshow(wordcloud, interpolation='bilinear')
      plt.axis('off')
      plt.title("Most Common Movie Genres")
      plt.show()
```

Most Common Movie Genres



```
[8]: # Check if both 'boxoffice' and 'genre' columns exist in the dataset.
if 'boxoffice' in joined_movies_dataset.columns and 'genre' in
    joined_movies_dataset.columns:

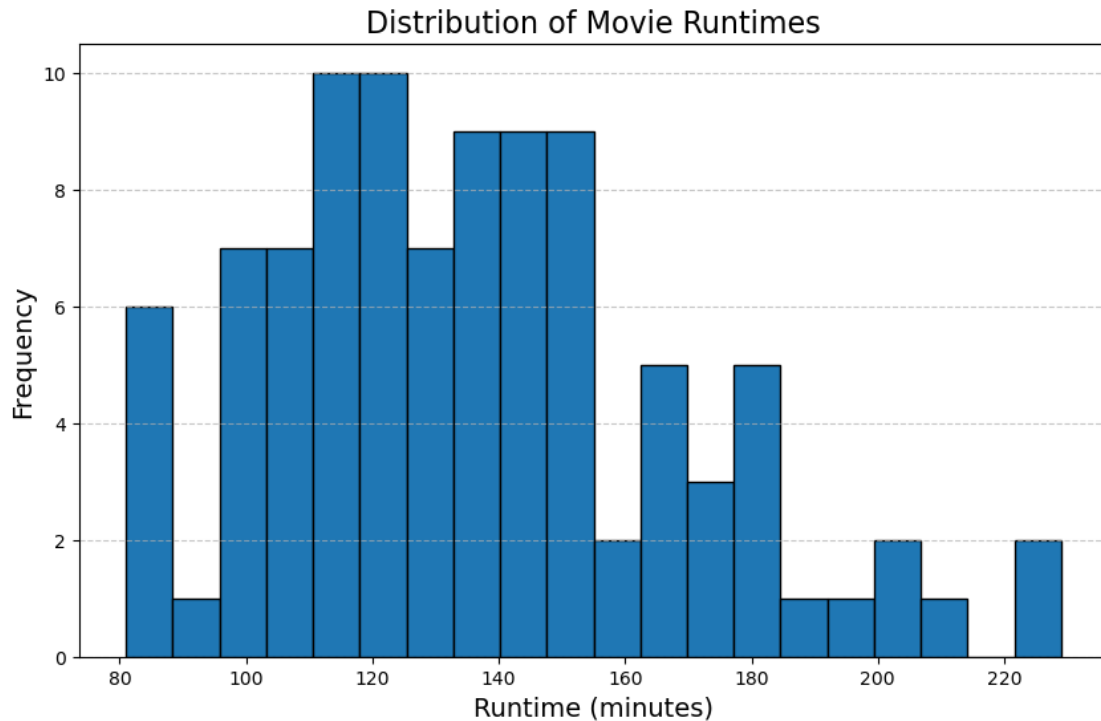
    # Group the data by 'genre', sum the 'boxoffice' revenue for each genre,
    # sort the values in descending order, and get the top 10 genres.
    boxoffice_by_genre = (joined_movies_dataset.groupby('genre')['boxoffice']
                           .sum()
                           .sort_values(ascending=False)
                           .head(10))

    plt.figure(figsize=(12, 6))
    boxoffice_by_genre.plot(kind='bar', color='green')
    plt.title('Top 10 Genres by Total Box Office Revenue', fontsize=16)
    plt.xlabel('Genre', fontsize=14)
    plt.ylabel('Total Box Office Revenue', fontsize=14)
    plt.xticks(rotation=45)
    plt.grid(axis='y', linestyle='--', alpha=0.7)
    plt.show()
```

```
[9]: # Check if the 'runtime' column exists in the dataset.
if 'runtime' in joined_movies_dataset.columns:
    plt.figure(figsize=(10, 6))

    # Plot the histogram of movie runtimes, excluding missing values (dropna).
    plt.hist(joined_movies_dataset['runtime'].dropna(), bins=20,
             edgecolor='black')
```

```
plt.title('Distribution of Movie Runtimes', fontsize=16)
plt.xlabel('Runtime (minutes)', fontsize=14)
plt.ylabel('Frequency', fontsize=14)
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.show()
```



1.0.1 Summary of Data Changes and Ethical Considerations

In this project, I worked with several datasets to analyze movie-related information. Here's a summary of the changes I made to the data, ethical considerations, and how I addressed them.

Changes Made to the Data

1. **Handling Missing Values:** Missing information in columns like `runtime` and `ratings` was either filled using averages or other logical methods or left out when it was unclear how to replace it.
2. **Outliers:** Extremely high or low values, like unrealistic movie runtimes, were removed to avoid skewing the analysis.
3. **Standardizing Data:** Columns like `movie titles` were cleaned to ensure consistency in capitalization and formatting. This helped when merging datasets.
4. **Feature Engineering:** New columns were created from existing data, such as extracting the release year from a date column.
5. **Encoding Data:** Text-based columns like `genres` and `content_rating` were converted into numerical formats to prepare for model building.

6. **Merging Data:** Data from multiple sources were combined using unique identifiers like IMDb ID to ensure the information was correctly aligned.

Legal and Regulatory Guidelines The data came from sources like [Rotten Tomatoes Movies and Critic Reviews Dataset](#), [IMDb Top 250 Movies](#), and [The Open Movie Database](#), all providing access under clear terms of use. I adhered to these policies, avoiding any unauthorized scraping or data extraction. No personal or sensitive information was included, ensuring compliance with data privacy laws like GDPR (General Data Protection Regulation).

Risks from Data Transformations Some risks emerged from the data changes:

- Imputed missing values might have introduced bias.
- Combining datasets could lead to errors if records didn't align correctly.
- Simplifying text data for numerical use could lose nuanced information.

Assumptions in Data Cleaning

- Missing values, like `runtime`, were assumed to follow existing data patterns.
- Columns across datasets, such as `ratings` and `genres`, referred to the same definitions.
- Data merging based on unique IDs was assumed to be accurate.

Data Credibility and Sourcing The datasets were obtained from trusted platforms like APIs, ensuring high-quality information. Cross-checking some values verified their credibility.

Ethical Considerations and Solutions The data was collected and processed ethically:

- Assumptions and changes were documented.
- Missing data handling was designed to minimize distortion.
- Validation checks ensured the accuracy of merging and transformations.
- Findings were interpreted with care, acknowledging limitations.

By being transparent and careful, I ensured the data transformations respected ethical standards and produced reliable, meaningful insights.

```
[10]: df1.head(60)
```

```
[10]:
```

	rank	title	release_year	\
0	1	The Shawshank Redemption	1994	
1	2	The Godfather	1972	
2	3	The Dark Knight	2008	
3	4	The Godfather Part II	1974	
4	5	12 Angry Men	1957	
5	6	The Lord of the Rings: The Return of the King	2003	
6	7	Schindler's List	1993	
7	8	Pulp Fiction	1994	
8	9	The Lord of the Rings: The Fellowship of the Ring	2001	
9	10	The Good, the Bad and the Ugly	1966	
10	11	Forrest Gump	1994	
11	12	The Lord of the Rings: The Two Towers	2002	

12	13	Fight Club	1999
13	14	Inception	2010
14	15	Star Wars: Episode V - The Empire Strikes Back	1980
15	16	The Matrix	1999
16	17	Goodfellas	1990
17	18	One Flew Over the Cuckoo's Nest	1975
18	19	Interstellar	2014
19	20	Se7en	1995
20	21	It's a Wonderful Life	1946
21	22	Seven Samurai	1954
22	23	The Silence of the Lambs	1991
23	24	Saving Private Ryan	1998
24	25	City of God	2002
25	26	The Green Mile	1999
26	27	Life Is Beautiful	1997
27	28	Terminator 2: Judgment Day	1991
28	29	Star Wars: Episode IV - A New Hope	1977
29	30	Back to the Future	1985
30	31	Spirited Away	2001
31	32	The Pianist	2002
32	33	Parasite	2019
33	34	Psycho	1960
34	35	Gladiator	2000
35	36	The Lion King	1994
36	37	The Departed	2006
37	38	Spider-Man: Across the Spider-Verse	2023
38	39	Whiplash	2014
39	40	Grave of the Fireflies	1988
40	41	American History X	1998
41	42	Léon: The Professional	1994
42	43	The Prestige	2006
43	44	Harakiri	1962
44	45	Casablanca	1942
45	46	The Usual Suspects	1995
46	47	The Intouchables	2011
47	48	Dune: Part Two	2024
48	49	Cinema Paradiso	1988
49	50	Modern Times	1936
50	51	Alien	1979
51	52	Rear Window	1954
52	53	Once Upon a Time in the West	1968
53	54	Django Unchained	2012
54	55	City Lights	1931
55	56	Apocalypse Now	1979
56	57	Memento	2000
57	58	WALL·E	2008
58	59	Raiders of the Lost Ark	1981

	content_rating	rating	runtime_minutes	rating_scaled
0	R	9.3	142	93.0
1	R	9.2	175	92.0
2	PG-13	9.0	152	90.0
3	R	9.0	202	90.0
4	Approved	9.0	96	90.0
5	PG-13	9.0	201	90.0
6	R	9.0	195	90.0
7	R	8.9	154	89.0
8	PG-13	8.9	178	89.0
9	R	8.8	178	88.0
10	PG-13	8.8	142	88.0
11	PG-13	8.8	179	88.0
12	R	8.8	139	88.0
13	PG-13	8.8	148	88.0
14	PG	8.7	124	87.0
15	R	8.7	136	87.0
16	R	8.7	145	87.0
17	R	8.7	133	87.0
18	PG-13	8.7	169	87.0
19	R	8.6	127	86.0
20	PG	8.6	130	86.0
21	Not Rated	8.6	207	86.0
22	R	8.6	118	86.0
23	R	8.6	169	86.0
24	R	8.6	130	86.0
25	R	8.6	189	86.0
26	PG-13	8.6	116	86.0
27	R	8.6	137	86.0
28	PG	8.6	121	86.0
29	PG	8.5	116	85.0
30	PG	8.6	124	86.0
31	R	8.5	150	85.0
32	R	8.5	132	85.0
33	R	8.5	109	85.0
34	R	8.5	155	85.0
35	G	8.5	88	85.0
36	R	8.5	151	85.0
37	PG	8.6	140	86.0
38	R	8.5	106	85.0
39	Not Rated	8.5	88	85.0
40	R	8.5	119	85.0
41	R	8.5	110	85.0
42	PG-13	8.5	130	85.0
43	Not Rated	8.6	133	86.0

44	PG	8.5	102	85.0
45	R	8.5	106	85.0
46	R	8.5	112	85.0
47	PG-13	8.5	166	85.0
48	PG	8.5	174	85.0
49	G	8.5	87	85.0
50	R	8.5	117	85.0
51	PG	8.5	112	85.0
52	PG-13	8.5	166	85.0
53	R	8.5	165	85.0
54	G	8.5	87	85.0
55	R	8.4	147	84.0
56	R	8.4	113	84.0
57	G	8.4	98	84.0
58	PG	8.4	115	84.0
59	R	8.4	137	84.0

```
[11]: df2.head(60)
```

```
[11]:
```

	title	year	rating	\
0	The Shawshank Redemption	1994	R	
1	The Godfather	1972	R	
2	The Dark Knight	2008	PG-13	
3	The Godfather Part II	1974	R	
4	12 Angry Men	1957	Approved	
5	The Lord of the Rings: The Return of the King	2003	PG-13	
6	Schindler's List	1994	R	
7	Pulp Fiction	1994	R	
8	The Lord of the Rings: The Fellowship of the Ring	2001	PG-13	
9	The Good, the Bad and the Ugly	1967	R	
10	Forrest Gump	1994	PG-13	
11	The Lord of the Rings: The Two Towers	2002	PG-13	
12	Fight Club	1999	R	
13	Inception	2010	PG-13	
14	Star Wars: Episode V - The Empire Strikes Back	1980	PG	
15	The Matrix	1999	R	
16	GoodFellas	1990	R	
17	One Flew Over the Cuckoo's Nest	1975	R	
18	Interstellar	2014	PG-13	
19	It's a Wonderful Life	1946	PG	
20	Seven Samurai	1954	Not Rated	
21	The Silence of the Lambs	1991	R	
22	Saving Private Ryan	1998	R	
23	City of God	2004	R	
24	Life Is Beautiful	1997	PG-13	
25	The Green Mile	1999	R	
26	Terminator 2: Judgment Day	1991	R	

27	Star Wars: Episode IV - A New Hope	1977	PG
28	Back to the Future	1985	PG
29	Spirited Away	2003	PG
30	The Pianist	2003	R
31	Parasite	2019	R
32	American Psycho	2000	R
33	Gladiator	2000	R
34	The Lion King	1994	G
35	Spider-Man: Across the Spider-Verse	2023	PG
36	The Departed	2006	R
37	Whiplash	2014	R
38	Grave of the Fireflies	1989	Not Rated
39	American History X	1998	R
40	Léon: The Professional	1994	R
41	The Prestige	2006	PG-13
42	Harakiri	1964	Not Rated
43	Casablanca	1943	PG
44	The Usual Suspects	1995	R
45	The Intouchables	2011	R
46	Dune: Part Two	2024	PG-13
47	Cinema Paradiso	1990	PG
48	Modern Times	1936	G
49	Alien	1979	R
50	Rear Window	1954	PG
51	Once Upon a Time in the West	1969	PG-13
52	Django Unchained	2012	R
53	City Lights	1931	G
54	Apocalypse Now	1979	R
55	Memento	2001	R
56	WALL·E	2008	G
57	Raiders of the Lost Ark	1981	PG
58	12th Fail	2023	NaN
59	The Lives of Others	2006	R

	release_date	runtime	genre \
0	1994-10-14	142.0	Drama
1	1972-03-24	175.0	Crime, Drama
2	2008-07-18	152.0	Action, Crime, Drama
3	1974-12-18	202.0	Crime, Drama
4	1957-04-10	96.0	Crime, Drama
5	2003-12-17	201.0	Adventure, Drama, Fantasy
6	1994-02-04	195.0	Biography, Drama, History
7	1994-10-14	154.0	Crime, Drama
8	2001-12-19	178.0	Adventure, Drama, Fantasy
9	1967-12-29	178.0	Adventure, Drama, Western
10	1994-07-06	142.0	Drama, Romance
11	2002-12-18	179.0	Adventure, Drama, Fantasy

12	1999-10-15	139.0	Crime, Drama, Thriller
13	2010-07-16	148.0	Action, Adventure, Sci-Fi
14	1980-06-18	124.0	Action, Adventure, Fantasy
15	1999-03-31	136.0	Action, Sci-Fi
16	1990-09-21	145.0	Biography, Crime, Drama
17	1975-11-21	133.0	Drama
18	2014-11-07	169.0	Adventure, Drama, Sci-Fi
19	1947-01-07	130.0	Drama, Family, Fantasy
20	1956-11-19	207.0	Action, Drama
21	1991-02-14	118.0	Crime, Drama, Horror
22	1998-07-24	169.0	Drama, War
23	2004-02-13	130.0	Crime, Drama
24	1997-12-20	116.0	Comedy, Drama, Romance
25	1999-12-10	189.0	Crime, Drama, Fantasy
26	1991-07-03	137.0	Action, Adventure, Sci-Fi
27	1977-05-25	121.0	Action, Adventure, Fantasy
28	1985-07-03	116.0	Adventure, Comedy, Sci-Fi
29	2003-03-28	124.0	Animation, Adventure, Family
30	2003-03-28	150.0	Biography, Drama, Music
31	2019-11-08	132.0	Drama, Thriller
32	2000-04-14	102.0	Crime, Drama, Horror
33	2000-05-05	155.0	Action, Adventure, Drama
34	1994-06-24	88.0	Animation, Adventure, Drama
35	2023-06-02	140.0	Animation, Action, Adventure
36	2006-10-06	151.0	Crime, Drama, Thriller
37	2014-10-15	106.0	Drama, Music
38	1989-07-26	88.0	Animation, Drama, War
39	1998-11-20	119.0	Crime, Drama
40	1994-11-18	110.0	Action, Crime, Drama
41	2006-10-20	130.0	Drama, Mystery, Sci-Fi
42	1964-08-04	133.0	Drama, Mystery
43	1943-01-23	102.0	Drama, Romance, War
44	1995-08-16	106.0	Crime, Drama, Mystery
45	2011-11-02	112.0	Comedy, Drama
46	2024-03-01	166.0	Action, Adventure, Drama
47	1990-02-23	174.0	Drama, Romance
48	1936-02-25	87.0	Comedy, Drama, Romance
49	1979-06-22	117.0	Horror, Sci-Fi
50	1954-09-01	112.0	Drama, Mystery, Thriller
51	1969-07-04	145.0	Drama, Western
52	2012-12-25	165.0	Drama, Western
53	1931-03-07	87.0	Comedy, Drama, Romance
54	1979-08-15	147.0	Drama, Mystery, War
55	2001-05-25	113.0	Drama, Mystery, Thriller
56	2008-06-27	98.0	Animation, Adventure, Family
57	1981-06-12	115.0	Action, Adventure
58	2023-10-27	147.0	Biography, Drama

59 2007-03-30 137.0 Drama, Mystery, Thriller

	director \
0	Frank Darabont
1	Francis Ford Coppola
2	Christopher Nolan
3	Francis Ford Coppola
4	Sidney Lumet
5	Peter Jackson
6	Steven Spielberg
7	Quentin Tarantino
8	Peter Jackson
9	Sergio Leone
10	Robert Zemeckis
11	Peter Jackson
12	David Fincher
13	Christopher Nolan
14	Irvin Kershner
15	Lana Wachowski, Lilly Wachowski
16	Martin Scorsese
17	Miloš Forman
18	Christopher Nolan
19	Frank Capra
20	Akira Kurosawa
21	Jonathan Demme
22	Steven Spielberg
23	Kátia Lund, Fernando Meirelles
24	Roberto Benigni
25	Frank Darabont
26	James Cameron
27	George Lucas
28	Robert Zemeckis
29	Hayao Miyazaki
30	Roman Polanski
31	Bong Joon Ho
32	Mary Harron
33	Ridley Scott
34	Roger Allers, Rob Minkoff
35	Joaquim Dos Santos, Kemp Powers, Justin K. Tho...
36	Martin Scorsese
37	Damien Chazelle
38	Isao Takahata
39	Tony Kaye
40	Luc Besson
41	Christopher Nolan
42	Masaki Kobayashi
43	Michael Curtiz

44 Bryan Singer
 45 Olivier Nakache, Éric Toledano
 46 Denis Villeneuve
 47 Giuseppe Tornatore
 48 Charles Chaplin
 49 Ridley Scott
 50 Alfred Hitchcock
 51 Sergio Leone
 52 Quentin Tarantino
 53 Charles Chaplin
 54 Francis Ford Coppola
 55 Christopher Nolan
 56 Andrew Stanton
 57 Steven Spielberg
 58 Vidhu Vinod Chopra
 59 Florian Henckel von Donnersmarck

writer \
 0 Stephen King, Frank Darabont
 1 Mario Puzo, Francis Ford Coppola
 2 Jonathan Nolan, Christopher Nolan, David S. Goyer
 3 Francis Ford Coppola, Mario Puzo
 4 Reginald Rose
 5 J.R.R. Tolkien, Fran Walsh, Philippa Boyens
 6 Thomas Keneally, Steven Zaillian
 7 Quentin Tarantino, Roger Avary
 8 J.R.R. Tolkien, Fran Walsh, Philippa Boyens
 9 Luciano Vincenzoni, Sergio Leone, Agenore Incr...
 10 Winston Groom, Eric Roth
 11 J.R.R. Tolkien, Fran Walsh, Philippa Boyens
 12 Chuck Palahniuk, Jim Uhls
 13 Christopher Nolan
 14 Leigh Brackett, Lawrence Kasdan, George Lucas
 15 Lilly Wachowski, Lana Wachowski
 16 Nicholas Pileggi, Martin Scorsese
 17 Lawrence Hauben, Bo Goldman, Ken Kesey
 18 Jonathan Nolan, Christopher Nolan
 19 Frances Goodrich, Albert Hackett, Frank Capra
 20 Akira Kurosawa, Shinobu Hashimoto, Hideo Oguni
 21 Thomas Harris, Ted Tally
 22 Robert Rodat
 23 Bráulio Mantovani, Paulo Lins
 24 Vincenzo Cerami, Roberto Benigni
 25 Stephen King, Frank Darabont
 26 James Cameron, William Wisher
 27 George Lucas
 28 Robert Zemeckis, Bob Gale

29 Hayao Miyazaki
 30 Ronald Harwood, Wladyslaw Szpilman
 31 Bong Joon Ho, Han Jin-won
 32 Bret Easton Ellis, Mary Harron, Guinevere Turner
 33 David Franzoni, John Logan, William Nicholson
 34 Irene Mecchi, Jonathan Roberts, Linda Woolverton
 35 Phil Lord, Christopher Miller, Dave Callaham
 36 William Monahan, Alan Mak, Felix Chong
 37 Damien Chazelle
 38 Akiyuki Nosaka, Isao Takahata
 39 David McKenna
 40 Luc Besson
 41 Jonathan Nolan, Christopher Nolan, Christopher...
 42 Yasuhiko Takiguchi, Shinobu Hashimoto
 43 Philip G. Epstein, Julius J. Epstein, Howard Koch
 44 Christopher McQuarrie
 45 Olivier Nakache, Éric Toledano, Philippe Pozzo...
 46 Denis Villeneuve, Jon Spaihts, Frank Herbert
 47 Giuseppe Tornatore, Vanna Paoli
 48 Charles Chaplin
 49 Dan O'Bannon, Ronald Shusett
 50 John Michael Hayes, Cornell Woolrich
 51 Sergio Donati, Sergio Leone, Dario Argento
 52 Quentin Tarantino
 53 Charles Chaplin, Harry Carr, Harry Crocker
 54 John Milius, Francis Ford Coppola, Michael Herr
 55 Christopher Nolan, Jonathan Nolan
 56 Andrew Stanton, Jim Reardon, Pete Docter
 57 Lawrence Kasdan, George Lucas, Philip Kaufman
 58 Vidhu Vinod Chopra, Jaskunwar Kohli, Anurag Pa...
 59 Florian Henckel von Donnersmarck

actors \

0 Tim Robbins, Morgan Freeman, Bob Gunton
 1 Marlon Brando, Al Pacino, James Caan
 2 Christian Bale, Heath Ledger, Aaron Eckhart
 3 Al Pacino, Robert De Niro, Robert Duvall
 4 Henry Fonda, Lee J. Cobb, Martin Balsam
 5 Elijah Wood, Viggo Mortensen, Ian McKellen
 6 Liam Neeson, Ralph Fiennes, Ben Kingsley
 7 John Travolta, Uma Thurman, Samuel L. Jackson
 8 Elijah Wood, Ian McKellen, Orlando Bloom
 9 Clint Eastwood, Eli Wallach, Lee Van Cleef
 10 Tom Hanks, Robin Wright, Gary Sinise
 11 Elijah Wood, Ian McKellen, Viggo Mortensen
 12 Brad Pitt, Edward Norton, Meat Loaf
 13 Leonardo DiCaprio, Joseph Gordon-Levitt, Ello...

14 Mark Hamill, Harrison Ford, Carrie Fisher
 15 Keanu Reeves, Laurence Fishburne, Carrie-Anne ...
 16 Robert De Niro, Ray Liotta, Joe Pesci
 17 Jack Nicholson, Louise Fletcher, Michael Berryman
 18 Matthew McConaughey, Anne Hathaway, Jessica Ch...
 19 James Stewart, Donna Reed, Lionel Barrymore
 20 Toshirô Mifune, Takashi Shimura, Keiko Tsushima
 21 Jodie Foster, Anthony Hopkins, Scott Glenn
 22 Tom Hanks, Matt Damon, Tom Sizemore
 23 Alexandre Rodrigues, Leandro Firmino, Matheus ...
 24 Roberto Benigni, Nicoletta Braschi, Giorgio Ca...
 25 Tom Hanks, Michael Clarke Duncan, David Morse
 26 Arnold Schwarzenegger, Linda Hamilton, Edward ...
 27 Mark Hamill, Harrison Ford, Carrie Fisher
 28 Michael J. Fox, Christopher Lloyd, Lea Thompson
 29 Miyu Irino, Rumi Hiiragi, Mari Natsuki
 30 Adrien Brody, Thomas Kretschmann, Frank Finlay
 31 Song Kang-ho, Lee Sun-kyun, Cho Yeo-jeong
 32 Christian Bale, Justin Theroux, Josh Lucas
 33 Russell Crowe, Joaquin Phoenix, Connie Nielsen
 34 Matthew Broderick, Jeremy Irons, James Earl Jones
 35 Shameik Moore, Hailee Steinfeld, Brian Tyree H...
 36 Leonardo DiCaprio, Matt Damon, Jack Nicholson
 37 Miles Teller, J.K. Simmons, Melissa Benoist
 38 Tsutomu Tatsumi, Ayano Shiraishi, Akemi Yamaguchi
 39 Edward Norton, Edward Furlong, Beverly D'Angelo
 40 Jean Reno, Gary Oldman, Natalie Portman
 41 Christian Bale, Hugh Jackman, Scarlett Johansson
 42 Tatsuya Nakadai, Akira Ishihama, Shima Iwashita
 43 Humphrey Bogart, Ingrid Bergman, Paul Henreid
 44 Kevin Spacey, Gabriel Byrne, Chazz Palminteri
 45 François Cluzet, Omar Sy, Anne Le Ny
 46 Timothée Chalamet, Zendaya, Rebecca Ferguson
 47 Philippe Noiret, Enzo Cannavale, Antonella Attili
 48 Charles Chaplin, Paulette Goddard, Henry Bergman
 49 Sigourney Weaver, Tom Skerritt, John Hurt
 50 James Stewart, Grace Kelly, Wendell Corey
 51 Henry Fonda, Charles Bronson, Claudia Cardinale
 52 Jamie Foxx, Christoph Waltz, Leonardo DiCaprio
 53 Charles Chaplin, Virginia Cherrill, Florence Lee
 54 Martin Sheen, Marlon Brando, Robert Duvall
 55 Guy Pearce, Carrie-Anne Moss, Joe Pantoliano
 56 Ben Burt, Elissa Knight, Jeff Garlin
 57 Harrison Ford, Karen Allen, Paul Freeman
 58 Vikrant Massey, Medha Shankr, Anant Joshi
 59 Ulrich Mühe, Martina Gedeck, Sebastian Koch

plot \

0 A wrongfully convicted banker forms a close fr...
1 The aging patriarch of an organized crime dyna...
2 When a menace known as the Joker wreaks havoc ...
3 The early life and career of Vito Corleone in ...
4 The jury in a New York City murder trial is fr...
5 Gandalf and Aragorn lead the World of Men agai...
6 In German-occupied Poland during World War II,...
7 The lives of two mob hitmen, a boxer, a gangst...
8 A meek Hobbit from the Shire and eight compani...
9 A bounty-hunting scam joins two men in an unea...
10 The history of the United States from the 1950...
11 While Frodo and Sam edge closer to Mordor with...
12 An insomniac office worker and a devil-may-car...
13 A thief who steals corporate secrets through t...
14 After the Empire overpowers the Rebel Alliance...
15 When a beautiful stranger leads computer hacke...
16 The story of Henry Hill and his life in the ma...
17 A rebellious convict is sent to a psychiatric ...
18 When Earth becomes uninhabitable in the future...
19 An angel is sent from Heaven to help a despera...
20 Farmers from a village exploited by bandits hi...
21 A young F.B.I. cadet must receive the help of ...
22 Following the Normandy Landings, a group of U...
23 In the slums of Rio, two kids' paths diverge a...
24 When an open-minded Jewish waiter and his son ...
25 A death row guard learns that a gentle giant i...
26 A cyborg from the future, identical to the one...
27 Luke Skywalker joins forces with a Jedi Knight...
28 Marty McFly, a 17-year-old high school student...
29 During her family's move to the suburbs, a sul...
30 During WWII, acclaimed Polish musician Wladysl...
31 Greed and class discrimination threaten the ne...
32 A wealthy New York City investment banking exe...
33 A former Roman General sets out to exact venge...
34 Lion prince Simba and his father are targeted ...
35 Traveling across the multiverse, Miles Morales...
36 An undercover cop and a mole in the police att...
37 A promising young drummer enrolls at a cut-thr...
38 A young boy and his little sister struggle to ...
39 A European-American man is released from priso...
40 When her family is murdered by a group of corr...
41 Two friends and fellow magicians become bitter...
42 When a ronin requesting seppuku at a feudal lo...
43 A cynical expatriate American cafe owner strug...
44 The sole survivor of a pier shoot-out tells th...
45 After he becomes a quadriplegic from a paragli...

46 Paul Atreides unites with the Fremmen while on ...
 47 Salvatore, a famous film director, returns to ...
 48 The Tramp struggles to live in modern industri...
 49 After investigating a mysterious transmission ...
 50 A bored photographer recovering from a broken ...
 51 A mysterious stranger with a harmonica teams u...
 52 With the help of a German bounty-hunter, a fre...
 53 With the aid of a wealthy erratic tippler, a d...
 54 A U.S. Army officer serving in Vietnam is task...
 55 A former insurance investigator who now suffer...
 56 A robot who is responsible for cleaning a wast...
 57 In 1936, archaeologist Indiana Jones is tasked...
 58 The real-life story of IPS Officer Manoj Kumar...
 59 In 1984 East Berlin, an agent of the secret po...

	Country \
0	United States
1	United States
2	United States, United Kingdom
3	United States
4	United States
5	New Zealand, United States
6	United States
7	United States
8	New Zealand, United States
9	Italy, Spain, West Germany, United States
10	United States
11	New Zealand, United States
12	United States, Germany
13	United Kingdom, United States
14	United States
15	United States, Australia
16	United States
17	United States
18	United States, United Kingdom, Canada
19	United States
20	Japan
21	United States
22	United States
23	Brazil
24	Italy
25	United States
26	United States, France
27	United States
28	United States
29	Japan, United States
30	France, Poland, Germany, United Kingdom, Unite...

31 South Korea
 32 United States
 33 United States, United Kingdom, Malta, Morocco
 34 United States
 35 United States
 36 United States, Hong Kong
 37 United States
 38 Japan
 39 United States
 40 France, United States
 41 United Kingdom, United States
 42 Japan
 43 United States
 44 United States, Germany
 45 France
 46 United States, Canada, United Arab Emirates, H...
 47 Italy, France
 48 United States
 49 United Kingdom, United States
 50 United States
 51 Italy, United States
 52 United States
 53 United States
 54 United States
 55 United States
 56 United States, Japan
 57 United States
 58 India
 59 Germany, France

	Awards	Metascore	imdbRating \
0	Nominated for 7 Oscars. 21 wins & 42 nominatio...	82.0	9.3
1	Won 3 Oscars. 31 wins & 31 nominations total	100.0	9.2
2	Won 2 Oscars. 163 wins & 165 nominations total	85.0	9.1
3	Won 6 Oscars. 17 wins & 21 nominations total	90.0	9.0
4	Nominated for 3 Oscars. 16 wins & 12 nominatio...	97.0	9.0
5	Won 11 Oscars. 215 wins & 124 nominations total	94.0	9.0
6	Won 7 Oscars. 91 wins & 49 nominations total	95.0	9.0
7	Won 1 Oscar. 69 wins & 72 nominations total	95.0	8.8
8	Won 4 Oscars. 126 wins & 127 nominations total	92.0	8.9
9	2 wins & 5 nominations total	90.0	8.8
10	Won 6 Oscars. 51 wins & 74 nominations total	82.0	8.8
11	Won 2 Oscars. 132 wins & 139 nominations total	87.0	8.8
12	Nominated for 1 Oscar. 12 wins & 38 nomination...	67.0	8.8
13	Won 4 Oscars. 160 wins & 220 nominations total	74.0	8.8
14	Won 1 Oscar. 27 wins & 21 nominations total	82.0	8.7
15	Won 4 Oscars. 42 wins & 52 nominations total	73.0	8.7

16	Won 1 Oscar. 44 wins & 38 nominations total	92.0	8.7
17	Won 5 Oscars. 38 wins & 15 nominations total	84.0	8.6
18	Won 1 Oscar. 45 wins & 148 nominations total	74.0	8.7
19	Nominated for 5 Oscars. 11 wins & 7 nomination...	89.0	8.6
20	Nominated for 2 Oscars. 5 wins & 9 nominations...	98.0	8.6
21	Won 5 Oscars. 71 wins & 50 nominations total	86.0	8.6
22	Won 5 Oscars. 79 wins & 75 nominations total	91.0	8.6
23	Nominated for 4 Oscars. 76 wins & 50 nominatio...	79.0	8.6
24	Won 3 Oscars. 72 wins & 52 nominations total	58.0	8.6
25	Nominated for 4 Oscars. 15 wins & 37 nominatio...	61.0	8.6
26	Won 4 Oscars. 39 wins & 33 nominations total	75.0	8.6
27	Won 6 Oscars. 70 wins & 31 nominations total	90.0	8.6
28	Won 1 Oscar. 27 wins & 25 nominations total	88.0	8.5
29	Won 1 Oscar. 58 wins & 31 nominations total	96.0	8.6
30	Won 3 Oscars. 57 wins & 74 nominations total	85.0	8.5
31	Won 4 Oscars. 309 wins & 261 nominations total	97.0	8.5
32	7 wins & 11 nominations total	64.0	7.6
33	Won 5 Oscars. 61 wins & 105 nominations total	67.0	8.5
34	Won 2 Oscars. 43 wins & 35 nominations total	88.0	8.5
35	Nominated for 1 Oscar. 107 wins & 164 nominati...	86.0	8.5
36	Won 4 Oscars. 100 wins & 141 nominations total	85.0	8.5
37	Won 3 Oscars. 100 wins & 144 nominations total	89.0	8.5
38	3 wins total	94.0	8.5
39	Nominated for 1 Oscar. 4 wins & 15 nominations...	62.0	8.5
40	5 wins & 16 nominations total	64.0	8.5
41	Nominated for 2 Oscars. 6 wins & 44 nomination...	66.0	8.5
42	9 wins & 3 nominations total	85.0	8.6
43	Won 3 Oscars. 18 wins & 12 nominations total	100.0	8.5
44	Won 2 Oscars. 37 wins & 17 nominations total	76.0	8.5
45	Nominated for 1 BAFTA Award. 38 wins & 40 nomi...	57.0	8.5
46	Won 2 Oscars. 125 wins & 376 nominations total	79.0	8.4
47	Won 1 Oscar. 25 wins & 33 nominations total	80.0	8.5
48	6 wins & 1 nomination total	96.0	8.5
49	Won 1 Oscar. 19 wins & 22 nominations total	89.0	8.5
50	Nominated for 4 Oscars. 7 wins & 14 nomination...	100.0	8.5
51	6 wins & 5 nominations total	82.0	8.5
52	Won 2 Oscars. 58 wins & 158 nominations total	81.0	8.5
53	6 wins total	99.0	8.5
54	Won 2 Oscars. 21 wins & 33 nominations total	94.0	8.4
55	Nominated for 2 Oscars. 57 wins & 59 nominatio...	83.0	8.4
56	Won 1 Oscar. 96 wins & 95 nominations total	95.0	8.4
57	Won 4 Oscars. 41 wins & 24 nominations total	86.0	8.4
58	31 wins & 40 nominations total	NaN	8.7
59	Won 1 Oscar. 80 wins & 38 nominations total	89.0	8.4

	imdbVotes	imdbID	BoxOffice	rt_rating
0	3,182,645	tt0111161	\$28,767,189	89%

1	2,222,804	tt0068646	\$136,381,073	97%
2	3,161,907	tt0468569	\$534,987,076	94%
3	1,492,120	tt0071562	\$47,834,595	96%
4	981,488	tt0050083	NaN	100%
5	2,161,991	tt0167260	\$386,975,644	94%
6	1,582,378	tt0108052	\$96,898,818	98%
7	2,431,656	tt0110912	\$107,928,762	92%
8	2,203,785	tt0120737	\$325,873,309	91%
9	889,734	tt0060196	\$25,100,000	97%
10	2,493,306	tt0109830	\$330,455,270	75%
11	1,954,618	tt0167261	\$350,104,956	95%
12	2,601,857	tt0137523	\$37,834,891	82%
13	2,811,614	tt1375666	\$292,587,330	87%
14	1,495,152	tt0080684	\$292,753,960	93%
15	2,243,093	tt0133093	\$177,559,005	83%
16	1,388,009	tt0099685	\$46,909,721	93%
17	1,155,304	tt0073486	\$108,981,275	92%
18	2,516,752	tt0816692	\$203,227,580	73%
19	552,062	tt0038650	\$1,483,643	94%
20	397,770	tt0047478	\$820,278	100%
21	1,718,588	tt0102926	\$130,742,922	95%
22	1,635,552	tt0120815	\$217,049,603	94%
23	869,206	tt0317248	\$7,564,459	91%
24	811,130	tt0118799	\$57,563,264	80%
25	1,557,536	tt0120689	\$136,801,374	78%
26	1,282,391	tt0103064	\$205,881,154	90%
27	1,555,851	tt0076759	\$460,998,507	94%
28	1,450,794	tt0088763	\$222,476,852	93%
29	964,088	tt0245429	\$14,821,254	96%
30	1,010,154	tt0253474	\$32,590,750	95%
31	1,148,210	tt6751668	\$53,847,897	99%
32	827,026	tt0144084	\$15,070,285	69%
33	1,844,518	tt0172495	\$187,705,427	80%
34	1,248,136	tt0110357	\$424,979,720	92%
35	509,515	tt9362722	\$381,593,754	95%
36	1,551,506	tt0407887	\$132,399,394	91%
37	1,134,459	tt2582802	\$14,003,391	94%
38	383,582	tt0095327	\$516,962	100%
39	1,271,167	tt0120586	\$6,719,864	84%
40	1,348,574	tt0110413	\$19,501,238	76%
41	1,583,310	tt0482571	\$53,089,891	77%
42	90,669	tt0056058	NaN	100%
43	654,681	tt0034583	\$4,219,709	99%
44	1,236,455	tt0114814	\$23,341,568	87%
45	1,024,790	tt1675434	\$10,198,820	75%
46	734,902	tt15239678	\$282,144,358	92%
47	316,943	tt0095765	\$12,397,210	90%

48	276,967	tt0027977	\$163,577	98%
49	1,061,640	tt0078748	\$84,206,106	93%
50	568,382	tt0047396	\$37,622,343	99%
51	383,734	tt0064116	\$5,321,508	96%
52	1,878,015	tt1853728	\$162,805,434	87%
53	215,345	tt0021749	\$19,181	95%
54	769,246	tt0078788	\$96,074,376	91%
55	1,432,382	tt0209144	\$25,544,867	93%
56	1,306,354	tt0910970	\$223,808,164	95%
57	1,102,713	tt0082971	\$248,159,971	94%
58	158,786	tt23849204	NaN	91%
59	440,932	tt0405094	\$11,286,112	92%

```
[12]: df3.head(60)
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[12]:
      Rotten Tomatoes Link \
0          m/0814255
1          m/0878835
2          m/10
3          m/1000013-12_angry_men
4    m/1000079-20000_leagues_under_the_sea
5          m/10000_bc
6          m/1000121-39_steps
7          m/1000123-310_to_yuma
8          m/10002008-charly
9    m/1000204-abraham_lincoln
10         m/10002114-dark_water
11         m/1000224-accused
12         m/10002516-lost_city
13    m/10002519-breaking_point
14         m/1000253-adams_rib
15    m/10002635-bridge_of_san_luis_rey
16         m/10002673-prowler
17         m/10003276-criminal
18    m/1000334-adventures_of_mark_twain
19         m/10003437-deep_blue
20    m/1000355-adventures_of_robin_hood
21         m/10003876-man_hunt
22         m/10003925-dead_end
23         m/10004050-bandwagon
24         m/10004069-whore
25    m/10004209-tristan_and_isolde
26         m/10004226-wild_side
27    m/10004232-10004232-tale_of_two_pizzas
28         m/10004288-running_scared
29    m/10004352-going_the_distance
30         m/10004504-ultraviolet
```

31 m/10004635-home_of_the_brave
 32 m/10004659-arthur
 33 m/10004684-malevolence
 34 m/10004697-eternal
 35 m/10004835-farewell_to_arms
 36 m/10004890-narrow_margin
 37 m/10004925-matador
 38 m/10005057-smile
 39 m/10005104-intruder
 40 m/10005178-saint_ralph
 41 m/10005403-the_baxter
 42 m/10005499-oliver_twist
 43 m/1000560-alamo
 44 m/10005755-g
 45 m/1000580-alexander_the_great
 46 m/1000581-alfie
 47 m/10005819-10005819-neverwas
 48 m/10006007-the_second_chance
 49 m/10006057-moonlight
 50 m/1000617-aliens
 51 m/10006188-rise
 52 m/1000626-all_about_eve
 53 m/10006270
 54 m/10006324
 55 m/10006343-dirty
 56 m/10006370-stay_alive
 57 m/1000640-all_of_me
 58 m/1000643-all_quiet_on_the_western_front
 59 m/10006434-dreamland

	Movie Title \
0	Percy Jackson & the Olympians: The Lightning T...
1	Please Give
2	10
3	12 Angry Men (Twelve Angry Men)
4	20,000 Leagues Under The Sea
5	10,000 B.C.
6	The 39 Steps
7	3:10 to Yuma
8	Charly (A Heartbeat Away)
9	Abraham Lincoln
10	Dark Water
11	The Accused
12	The Lost City
13	The Breaking Point
14	Adam's Rib
15	The Bridge of San Luis Rey

16 The Prowler (Cost of Living)
 17 Criminal
 18 The Adventures of Mark Twain
 19 Deep Blue
 20 The Adventures of Robin Hood
 21 Man Hunt
 22 Dead End
 23 Bandwagon
 24 Whore
 25 Tristan & Isolde
 26 Wild Side
 27 A Tale of Two Pizzas
 28 Running Scared
 29 Going the Distance
 30 Ultraviolet
 31 Home of the Brave
 32 Arthur and the Invisibles (Arthur and the Mini...
 33 Malevolence
 34 Eternal
 35 A Farewell To Arms
 36 The Narrow Margin
 37 The Matador
 38 Smile
 39 The Intruder (L'Intrus)
 40 Saint Ralph
 41 The Baxter
 42 Oliver Twist
 43 The Alamo
 44 G
 45 Alexander the Great
 46 Alfie
 47 Neverwas
 48 The Second Chance
 49 Moonlight
 50 Aliens
 51 Rise: Blood Hunter
 52 All About Eve
 53 When Do We Eat?
 54 The Mostly Unfabulous Social Life of Ethan Green
 55 Dirty
 56 Stay Alive
 57 All of Me
 58 All Quiet on the Western Front
 59 Dreamland

Movie Info \

0 Always trouble-prone, the life of teenager Per...

1 Kate (Catherine Keener) and her husband Alex (...
 2 A successful, middle-aged Hollywood songwriter...
 3 Following the closing arguments in a murder tr...
 4 In 1866, Professor Pierre M. Aronnax (Paul Luk...
 5 Mammoth hunter D'Leh (Steven Strait) has long ...
 6 While on vacation in London, Canadian Richard ...
 7 Dan Evans (Van Heflin), a drought-plagued Ariz...
 8 Cultural differences, past loves and personal ...
 9 The 16th U.S. president (Walter Huston) is por...
 10 In this moody Japanese horror film, newly-sing...
 11 Out drinking one night after a fight with her ...
 12 Fico Fellove (Andy Garcia), an apolitical Hava...
 13 A charter-boat captain winds up in the middle ...
 14 A courtroom rivalry finds its way into the hou...
 15 During the Spanish Inquisition, Franciscan mon...
 16 After being frightened by a peeping Tom at her...
 17 Needing a new partner capable of intricate con...
 18 In this claymation film, celebrated author Mar...
 19 This nature documentary shows the lives of dif...
 20 When King Richard the Lionheart is captured, h...
 21 Renowned big-game hunter Captain Thorndike (Wa...
 22 When a family en route to a Christmas Eve gath...
 23 When Tony (Lee Holmes) loses his job, he decid...
 24 Encouraged by her neighbor (Daryl Hannah), a f...
 25 After the fall of the Roman Empire, English or...
 26 Along with one (Edouard Nikitine) of her live-...
 27 The offspring of rival pizzeria owners (Vincen...
 28 Mafia flunky Joey Gazelle (Paul Walker) is ord...
 29 Nick (Christopher Jacot) is a successful young...
 30 In the late 21st-century, a vampiric disease m...
 31 In 1965, Klansmen murder a civil-rights worker...
 32 Ten-year-old Arthur must find a way to keep th...
 33 Bank robbers (Brandon Johnson, Heather Magee) ...
 34 A detective's (Conrad Pla) search for his miss...
 35 Based on the Ernest Hemingway novel, this Worl...
 36 Tense story of a tough cop attempting to trans...
 37 Salesman Danny Wright (Greg Kinnear) is workin...
 38 While visiting China, an American high-school ...
 39 After receiving a heart transplant, an enigmat...
 40 The year 1954 finds teen outcast Ralph Walker ...
 41 Elliot Sherman (Michael Showalter) has had ter...
 42 An orphan (Barney Clark) in 19th-century Londo...
 43 In 1836, the Mexican army, led by General Sant...
 44 Summer G (Richard T. Jones), an aspiring rappe...
 45 The son of Philip of Macedonia (Fredric March)...
 46 Set in postwar London, "Alfie" features Michael...
 47 Dr. Zach Riley (Aaron Eckhart) begins practici...

48 Two pastors, one from a wealthy parish in a pr...
 49 Claire, a Dutch girl, discovers a wounded boy ...
 50 After floating in space for 57 years, Lt. Rippl...
 51 Sadie (Lucy Liu), a reporter, investigates an ...
 52 Backstage story revolving around aspiring actr...
 53 Members of the Stuckman family gather for Sede...
 54 A young man named Ethan (Daniel Letterle) is h...
 55 Armando Sancho (Clifton Collins Jr.) is a form...
 56 Teenagers decide to pass idle time by playing ...
 57 While on her deathbed, the rich Edwina Cutwate...
 58 At the start of World War I, Paul Baumer (Rich...
 59 Teenage Aubrey (Agnes Bruckner) is always ther...

	Critics Consensus	Content Rating \
0	Though it may seem like just another Harry Pot...	PG
1	Nicole Holofcener's newest might seem slight i...	R
2	Blake Edwards' bawdy comedy may not score a pe...	R
3	Sidney Lumet's feature debut is a superbly wri...	NR
4	One of Disney's finest live-action adventures,...	G
5	With attention strictly paid to style instead ...	PG-13
6	Packed with twists and turns, this essential e...	NR
7	No Consensus	NR
8	No Consensus	PG
9	No Consensus	NR
10	No Consensus	PG-13
11	No Consensus	R
12	Its heart is in the right place, but what star...	R
13	No Consensus	NR
14	Matched by Garson Kanin's witty, sophisticated...	NR
15	Despite an all-star cast and some impressive v...	PG
16	No Consensus	PG
17	If you saw Nine Queens, it may feel redundant,...	R
18	No Consensus	G
19	Full of visually striking footage, Deep Blue m...	G
20	Errol Flynn thrills as the legendary title cha...	PG
21	No Consensus	NR
22	No Consensus	R
23	No Consensus	R
24	No Consensus	R
25	Competent but somewhat static, Tristan & Isold...	PG-13
26	No Consensus	NR
27	No Consensus	PG
28	This film runs with frenetic energy punctuated...	R
29	No Consensus	R
30	An incomprehensible and forgettable sci-fi thr...	PG-13
31	No Consensus	R
32	Arthur wastes its big-name voice talent on a p...	PG

33	No Consensus	R
34	Even by the standards of camp, Eternal fails t...	R
35	No Consensus	NR
36	No Consensus	NR
37	This humorously amoral, oddball comic thriller...	R
38	No Consensus	PG-13
39	The impressionistic narrative may confound the...	G
40	Saint Ralph is an amiable tear-jerker that wal...	PG-13
41	The Baxter is good-natured, but there are simp...	PG-13
42	Polanski's version of Dickens' classic won't h...	PG-13
43	John Wayne proves to be less compelling behind...	NR
44	Race and class struggles come to the Hamptons ...	R
45	No Consensus	G
46	Anchored by Michael Caine's charmingly droll p...	PG
47	No Consensus	PG-13
48	No Consensus	PG-13
49	No Consensus	R
50	While Alien was a marvel of slow-building, atm...	R
51	No Consensus	R
52	Smart, sophisticated, and devastatingly funny,...	PG
53	A crass, sitcom-ish holiday comedy, When Do We...	R
54	Ethan Green makes for an annoying protagonist,...	R
55	Dirty is an unoriginal L.A. cop drama that wea...	R
56	A by-the-numbers teen horror flick, Stay Alive...	PG-13
57	A high-concept farce carried by Carl Reiner's ...	PG
58	No Consensus	NR
59	No Consensus	PG-13

	Genres \
0	Action & Adventure, Comedy, Drama, Science Fic...
1	Comedy
2	Comedy, Romance
3	Classics, Drama
4	Action & Adventure, Drama, Kids & Family
5	Action & Adventure, Classics, Drama
6	Action & Adventure, Classics, Mystery & Suspense
7	Classics, Drama, Western
8	Comedy, Drama, Romance
9	Classics, Drama
10	Art House & International, Horror, Mystery & S...
11	Drama, Mystery & Suspense
12	Drama
13	Drama
14	Classics, Comedy, Romance
15	Art House & International, Drama, Faith & Spir...
16	Drama, Mystery & Suspense
17	Drama, Mystery & Suspense

18 Action & Adventure, Animation, Kids & Family
 19 Documentary, Special Interest
 20 Action & Adventure, Classics, Romance
 21 Classics, Drama, Mystery & Suspense
 22 Comedy, Horror, Mystery & Suspense
 23 Comedy
 24 Art House & International, Drama
 25 Drama, Science Fiction & Fantasy, Romance
 26 Art House & International, Drama
 27 Comedy
 28 Action & Adventure, Drama, Mystery & Suspense
 29 Comedy
 30 Action & Adventure, Drama, Horror, Science Fic...
 31 Action & Adventure, Documentary, Drama, Specia...
 32 Animation, Kids & Family, Science Fiction & Fa...
 33 Horror
 34 Horror, Mystery & Suspense
 35 Classics, Drama, Romance
 36 Classics, Drama, Mystery & Suspense
 37 Action & Adventure, Comedy, Drama, Mystery & S...
 38 Art House & International, Comedy, Drama
 39 Art House & International, Drama
 40 Comedy, Drama
 41 Comedy, Romance
 42 Drama, Kids & Family
 43 Action & Adventure, Classics, Drama, Western
 44 Drama, Romance
 45 Classics
 46 Classics, Comedy, Drama, Romance
 47 Drama, Mystery & Suspense
 48 Drama, Faith & Spirituality
 49 Drama, Mystery & Suspense
 50 Action & Adventure, Horror, Science Fiction & ...
 51 Horror, Mystery & Suspense
 52 Classics, Drama
 53 Comedy
 54 Comedy, Drama
 55 Drama
 56 Horror, Mystery & Suspense
 57 Comedy, Science Fiction & Fantasy
 58 Drama
 59 Art House & International, Drama

Directors \

0 Chris Columbus
 1 Nicole Holofcener
 2 Blake Edwards

3	Sidney Lumet
4	Richard Fleischer
5	Roland Emmerich
6	Alfred Hitchcock
7	Delmer Daves
8	Adam Thomas Anderegg
9	D.W. Griffith
10	Hideo Nakata
11	Jonathan Kaplan
12	Andy Garcia
13	Michael Curtiz
14	George Cukor
15	Mary McGuckian
16	Joseph Losey
17	Greggory Jacobs, Gregory Jacobs, Greg Jacobs
18	Will Vinton
19	Andy Byatt, Alastair Fothergill
20	Michael Curtiz, William Keighley
21	Fritz Lang
22	Jean-Baptiste Andrea, Fabrice Canepa, Jean-Bap...
23	John Schultz
24	María Lidón
25	Kevin Reynolds
26	Sébastien Lifshitz
27	Vincent Sassone
28	Wayne Kramer
29	Mark Griffiths
30	Kurt Wimmer
31	Paola di Florio
32	Luc Besson
33	Stevan Mena
34	Wilhelm Liebenberg, Federico Sanchez
35	Charles Vidor, John Hudson, John Huston
36	Richard Fleischer
37	Bob Yari, Richard Shepard
38	Jeff Kramer, Jeffrey Kramer
39	Claire Denis
40	Michael McGowan
41	Michael Showalter
42	Roman Polanski
43	John Wayne
44	Christopher Scott Cherot
45	Robert Rossen
46	Lewis Gilbert
47	Joshua Michael Stern
48	Steve Taylor
49	Paula Van der Oest

50 James Cameron
 51 Sebastian Gutierrez
 52 Joseph L. Mankiewicz
 53 Salvador Litvak
 54 George Bamber
 55 Chris Fisher
 56 William Brent Bell
 57 Carl Reiner
 58 Delbert Mann
 59 Jason Matzner

Authors \

0 Craig Titley, Chris Columbus, Rick Riordan
 1 Nicole Holofcener
 2 Blake Edwards
 3 Reginald Rose
 4 Earl Felton
 5 Harald Kloser, Roland Emmerich
 6 Alma Reville, Charles Bennett, Ian Hay, Alfred...
 7 Halsted Welles
 8 Jack Weyland, Janine Whetton Gilbert
 9 Gerrit J. Lloyd, Stephen Vincent Benét
 10 Hideo Hakata, Takashige Ichise, Yoshihiro Naka...
 11 Tom Topor
 12 G. Cabrera Infante, Guillermo Cabrera Infante
 13 Ranald MacDougall
 14 Garson Kanin, Ruth Gordon
 15 Mary McGuckian
 16 Robert Thoeren, Hugo Butler, Dalton Trumbo
 17 Gregory Jacobs, Fabián Bielinsky, Steven Soder...
 18 Susan Shadburne
 19 Alastair Fothergill, Andy Byatt, Tim Ecott
 20 Norman Reilly Raine, Seton I. Miller, Michael ...
 21 Dudley Nichols
 22 Jean-Baptiste Andrea, Fabrice Canepa
 23 John Schultz
 24 Adela Ibañez, Isabel Pisano
 25 Dean Georgaris
 26 Sébastien Lifshitz, Stéphane Bouquet
 27 Vincent Sassone
 28 Wayne Kramer
 29 Eric Goodman, Kelly Senecal
 30 Kurt Wimmer
 31 Paola di Florio
 32 Luc Besson, Celine Garcia
 33 Stevan Mena
 34 Wilhelm Liebenberg, Federico Sanchez

35 Ben Hecht
 36 Earl Felton, Martin Goldsmith
 37 Richard Shepard, Richard Shepherd
 38 Jeff Kramer, Jeffrey Kramer
 39 Claire Denis, Jean-Pol Fargeau
 40 Michael McGowan
 41 Michael Showalter
 42 Ronald Harwood
 43 James Edward Grant
 44 Charles E. Drew, Christopher Scott Cherot, Cha...
 45 Robert Rossen
 46 Bill Naughton
 47 Joshua Michael Stern
 48 Steve Taylor, Chip Arnold, Ben Pearson, Henry ...
 49 Carel Donck
 50 James Cameron
 51 Sebastian Gutierrez
 52 Joseph L. Mankiewicz
 53 Salvador Litvak, Nina Davidovich
 54 David Vernon
 55 Chris Fisher, Gil Reavill, Eric Saks
 56 William Brent Bell, Matthew Peterman
 57 Henry Olek, Carl Reiner, Phil Alden Robinson
 58 Paul Monash
 59 Tom Willett, Jason Matzner

	Actors	Original Release Date \
0	Logan Lerman, Brandon T. Jackson, Alexandra Da...	2010-02-12
1	Catherine Keener, Amanda Peet, Oliver Platt, R...	2010-04-30
2	Dudley Moore, Bo Derek, Julie Andrews, Robert ...	1979-10-05
3	Martin Balsam, John Fiedler, Lee J. Cobb, E.G...	1957-04-13
4	James Mason, Kirk Douglas, Paul Lukas, Peter L...	1954-01-01
5	Steven Strait, Camilla Belle, Cliff Curtis, Jo...	2008-03-07
6	Robert Donat, Madeleine Carroll, Godfrey Tearl...	1935-08-01
7	Glenn Ford, Van Heflin, Felicia Farr, Leora Da...	1957-08-07
8	Heather Beers, Gary Neilson, Lisa McCammon, Ja...	2002-09-27
9	Walter Huston, Una Merkel, Kay Hammond, Ian Ke...	1930-11-08
10	Hitomi Kuroki, Rio Kanno, Shigemitsu Ogi, Mire...	2002-01-19
11	Jodie Foster, Kelly McGillis, Bernie Coulson, ...	1988-10-14
12	Andy Garcia, Dustin Hoffman, Bill Murray, Inés...	2005-09-03
13	John Garfield, Patricia Neal, Phyllis Thaxter,...	1950-10-06
14	Spencer Tracy, Katharine Hepburn, Judy Hollida...	1949-11-18
15	Gabriel Byrne, F. Murray Abraham, Kathy Bates,...	2005-06-10
16	Van Heflin, Evelyn Keyes, Katherine Warren, Jo...	1951-12-03
17	John C. Reilly, Diego Luna, Maggie Gyllenhaal,...	2004-09-24
18	James Whitmore, Michele Mariana, Gary Krug, Ch...	1985-03-01
19	Pierce Brosnan, Michael Gambon	2005-06-17

20	Errol Flynn, Olivia de Havilland, Basil Rathbo...	1938-05-14
21	Walter Pidgeon, Joan Bennett, George Sanders, ...	1941-01-01
22	Ray Wise, Lin Shaye, Mick Cain, Alexandra Hold...	2003-09-26
23	Matthew Hennessey, Kevin Corrigan, Steve Parla...	1997-09-12
24	Daryl Hannah, Denise Richards, Joaquim de Alme...	2004-04-16
25	James Franco, Sophia Myles, Rufus Sewell, Davi...	2006-01-13
26	Stéphanie Michelini, Josiane Stoleru, Yasmine ...	2005-06-10
27	Vincent Pastore, Vincent Sassone, Frank Vincen...	2003-11-07
28	Paul Walker, Cameron Bright, Vera Farmiga, Kar...	2006-02-24
29	Christopher Jacot, Shawn Roberts, Ryan Bellevi...	2004-08-20
30	Milla Jovovich, Cameron Bright, Nick Chinlund,...	2006-03-03
31	Mary Liuzzo, Tony Liuzzo, Sally Liuzzo, Penny ...	2004-05-11
32	Freddie Highmore, Mia Farrow, Penny Balfour, D...	2007-01-12
33	Samantha Dark, R. Brandon Johnson, Heather Mag...	2004-09-10
34	Sarah Manninen, Caroline Neron, Victoria Sanch...	2004-08-26
35	Rock Hudson, Jennifer Jones, Vittorio De Sica,...	1957-12-14
36	Charlie MacGraw, Charles McGraw, Marie Windsor...	1952-05-04
37	Pierce Brosnan, Greg Kinnear, Hope Davis, Phil...	2006-01-27
38	Mika Boorem, Wang Luoyong, Yi Ding (II), Beau ...	2005-04-08
39	Beatrice Dalle, Michael Subor, Jean-Marc Terii...	2005-12-23
40	Adam Butcher, Campbell Scott, Gordon Pinsent, ...	2005-08-05
41	Michael Showalter, Elizabeth Banks, Michelle W...	2005-08-26
42	Ben Kingsley, Barney Clark, Leanne Rowe, Mark ...	2005-09-30
43	John Wayne, Richard Widmark, Laurence Harvey, ...	1960-10-24
44	Richard T. Jones, Blair Underwood, Chenoa Maxw...	2005-09-16
45	Richard Burton, Fredric March, Claire Bloom, D...	1956-03-28
46	Michael Caine, Millicent Martin, Julia Foster,...	1966-08-24
47	Aaron Eckhart, Ian McKellen, Brittany Murphy, ...	2005-09-09
48	Michael W. Smith, Jeff Obafemi Carr, Jonathan ...	2006-02-17
49	Laurien Van Den Broeck, Hunter Bussemaker, Joh...	2003-10-17
50	Sigourney Weaver, Carrie Henn, Michael Biehn, ...	1986-07-18
51	Lucy Liu, Michael Chiklis, Carla Gugino, James...	2007-06-01
52	Bette Davis, Anne Baxter, Gary Merrill, George...	1950-01-01
53	Michael Lerner, Lesley Ann Warren, Jack Klugma...	2006-04-07
54	Daniel Letterle, Meredith Baxter, David Monaha...	2006-06-16
55	Cuba Gooding Jr., Wyclef Jean, Keith David, Cl...	2006-02-24
56	Jon Foster, Frankie Muniz, Samaire Armstrong, ...	2006-03-24
57	Steve Martin, Lily Tomlin, Victoria Tennant, M...	1984-09-21
58	Richard Thomas, Ernest Borgnine, Ian Holm, Don...	1800-01-01
59	Agnes Bruckner, Kelli Garner, Justin Long, Gin...	2006-12-19

	...	Audience Status	Audience Rating	Audience Count	\
0	...	Spilled	53.0	254421.0	
1	...	Upright	64.0	11574.0	
2	...	Spilled	53.0	14684.0	
3	...	Upright	97.0	105386.0	
4	...	Upright	74.0	68918.0	

5	...	Spilled	37.0	411140.0
6	...	Upright	86.0	23890.0
7	...	Upright	79.0	9243.0
8	...	Upright	87.0	4819.0
9	...	Spilled	40.0	457.0
10	...	Upright	66.0	21475.0
11	...	Upright	79.0	20821.0
12	...	Upright	64.0	25944.0
13	...	Upright	86.0	335.0
14	...	Upright	86.0	10563.0
15	...	Spilled	35.0	1935.0
16	...	Upright	86.0	286.0
17	...	Spilled	57.0	6711.0
18	...	Upright	82.0	1046.0
19	...	Upright	80.0	4692.0
20	...	Upright	89.0	33946.0
21	...	Upright	75.0	554.0
22	...	Upright	60.0	10373.0
23	...	Upright	91.0	386.0
24	...	Upright	63.0	914.0
25	...	Upright	71.0	220956.0
26	...	Upright	80.0	867.0
27	...	Unknown Audience Status	63.0	4277.0
28	...	Upright	79.0	135400.0
29	...	Upright	61.0	13033.0
30	...	Spilled	30.0	279994.0
31	...	Spilled	48.0	866.0
32	...	Spilled	56.0	134483.0
33	...	Upright	83.0	2622.0
34	...	Spilled	39.0	1295.0
35	...	Spilled	37.0	1460.0
36	...	Upright	88.0	1135.0
37	...	Upright	65.0	73701.0
38	...	Upright	66.0	2059.0
39	...	Upright	73.0	9701.0
40	...	Upright	85.0	6203.0
41	...	Upright	75.0	7436.0
42	...	Upright	63.0	23294.0
43	...	Upright	63.0	7337.0
44	...	Upright	60.0	1143.0
45	...	Spilled	42.0	5383.0
46	...	Upright	76.0	14669.0
47	...	Upright	65.0	4059.0
48	...	Upright	78.0	79542.0
49	...	Upright	67.0	1430.0
50	...	Upright	94.0	429674.0
51	...	Spilled	20.0	8239.0

52	...	Upright	94.0	45248.0
53	...	Upright	68.0	1121.0
54	...	Spilled	58.0	2394.0
55	...	Spilled	42.0	7326.0
56	...	Spilled	55.0	113424.0
57	...	Upright	67.0	14346.0
58	...	Upright	69.0	7588.0
59	...	Upright	65.0	5650.0

	Tomatometer Top Critics Count	Tomatometer Fresh Critics Count \
0	43	73
1	44	123
2	2	16
3	6	54
4	5	24
5	37	12
6	8	49
7	6	27
8	0	2
9	4	9
10	3	12
11	1	20
12	37	21
13	1	10
14	6	27
15	13	1
16	6	18
17	35	85
18	1	4
19	18	35
20	8	48
21	0	11
22	1	6
23	1	4
24	4	0
25	36	38
26	4	10
27	5	3
28	39	54
29	2	0
30	21	7
31	7	14
32	30	20
33	4	6
34	10	7
35	6	0
36	0	11

37	39	117
38	6	2
39	10	26
40	26	44
41	25	24
42	39	87
43	4	11
44	17	7
45	1	0
46	6	26
47	1	1
48	8	7
49	6	6
50	17	74
51	2	5
52	22	76
53	19	16
54	13	8
55	10	6
56	18	6
57	10	35
58	0	5
59	5	10

	Tomatometer	Rotten Critics Count	z_score_Tomatometer Rating \
0		76	-0.422902
1		19	0.914255
2		8	0.210488
3		0	1.371703
4		3	0.984632
5		137	-1.865624
6		2	1.230950
7		1	1.230950
8		8	-1.443364
9		2	0.738313
10		3	0.667937
11		2	1.055008
12		62	-1.267422
13		0	1.371703
14		1	1.230950
15		23	-2.006377
16		0	1.371703
17		39	0.280865
18		1	0.667937
19		17	0.210488
20		0	1.371703
21		1	1.090197

22	2	0.491995
23	1	0.667937
24	6	-2.147130
25	83	-1.056292
26	6	0.069735
27	5	-0.809974
28	77	-0.704409
29	5	-2.147130
30	76	-1.865624
31	1	1.125385
32	72	-1.372987
33	12	-0.985915
34	27	-1.408175
35	12	-2.147130
36	0	1.371703
37	39	0.491995
38	11	-1.619305
39	4	0.914255
40	25	0.104923
41	51	-1.021104
42	56	-0.000642
43	10	-0.317337
44	22	-1.302610
45	8	-2.147130
46	1	1.230950
47	6	-1.654494
48	9	-0.598844
49	4	-0.035830
50	2	1.266138
51	9	-0.880350
52	1	1.336515
53	22	-0.669220
54	19	-1.091480
55	22	-1.408175
56	53	-1.795247
57	6	0.843878
58	0	1.371703
59	5	0.210488

	z_score_Audience Rating	z_score_Tomatometer Count	z_score_Audience Count
0	-0.378467	1.382400	0.073007
1	0.161389	1.278319	-0.072841
2	-0.378467	-0.476184	-0.070973
3	1.780959	-0.030124	-0.016500
4	0.652168	-0.431578	-0.038402
5	-1.163713	1.382400	0.167128
6	1.241102	-0.074730	-0.065444

7	0.897557	-0.416710	-0.074241
8	1.290180	-0.684346	-0.076898
9	-1.016479	-0.669477	-0.079518
10	0.259545	-0.610003	-0.066895
11	0.897557	-0.505922	-0.067288
12	0.161389	0.401068	-0.064211
13	1.241102	-0.684346	-0.079591
14	1.241102	-0.416710	-0.073448
15	-1.261868	-0.476184	-0.078630
16	1.241102	-0.565397	-0.079620
17	-0.182156	1.010683	-0.075762
18	1.044791	-0.758689	-0.079164
19	0.946635	-0.059862	-0.076974
20	1.388336	-0.119336	-0.059405
21	0.701246	-0.654609	-0.079459
22	-0.034922	-0.714083	-0.073562
23	1.486492	-0.758689	-0.079560
24	0.112312	-0.743821	-0.079243
25	0.504934	0.966077	0.052909
26	0.946635	-0.595134	-0.079271
27	0.112312	-0.714083	-0.077223
28	0.897557	1.114764	0.001526
29	0.014156	-0.758689	-0.071965
30	-1.507258	0.401068	0.088365
31	-0.623856	-0.610003	-0.079272
32	-0.231233	0.534886	0.000975
33	1.093869	-0.565397	-0.078217
34	-1.065557	-0.327498	-0.079014
35	-1.163713	-0.654609	-0.078915
36	1.339258	-0.669477	-0.079110
37	0.210467	1.486481	-0.035529
38	0.259545	-0.639740	-0.078556
39	0.603090	-0.386972	-0.073966
40	1.192024	0.192906	-0.076067
41	0.701246	0.282118	-0.075326
42	0.112312	1.293188	-0.065802
43	0.112312	-0.520791	-0.075386
44	-0.034922	-0.401841	-0.079106
45	-0.918323	-0.714083	-0.076559
46	0.750324	-0.431578	-0.070982
47	0.210467	-0.728952	-0.077354
48	0.848479	-0.595134	-0.032021
49	0.308623	-0.684346	-0.078933
50	1.633725	0.296987	0.178260
51	-1.998036	-0.624871	-0.074844
52	1.633725	0.311855	-0.052617
53	0.357701	-0.268023	-0.079119

54	-0.133078	-0.431578	-0.078354
55	-0.918323	-0.416710	-0.075392
56	-0.280311	0.044219	-0.011672
57	0.308623	-0.223417	-0.071176
58	0.406779	-0.758689	-0.075235
59	0.210467	-0.610003	-0.076399

[60 rows x 26 columns]

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[13]: joined_df.head(60)
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[13]:
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	rank	title	release_year	\
0	1	The Shawshank Redemption	1994	
1	2	The Godfather	1972	
2	3	The Dark Knight	2008	
3	4	The Godfather Part II	1974	
4	5	12 Angry Men	1957	
5	6	The Lord of the Rings: The Return of the King	2003	
6	7	Schindler's List	1993	
7	8	Pulp Fiction	1994	
8	9	The Lord of the Rings: The Fellowship of the Ring	2001	
9	10	The Good, the Bad and the Ugly	1966	
10	11	Forrest Gump	1994	
11	12	The Lord of the Rings: The Two Towers	2002	
12	13	Fight Club	1999	
13	14	Inception	2010	
14	15	Star Wars: Episode V - The Empire Strikes Back	1980	
15	16	The Matrix	1999	
16	17	Goodfellas	1990	
17	18	One Flew Over the Cuckoo's Nest	1975	
18	19	Interstellar	2014	
19	20	Se7en	1995	
20	21	It's a Wonderful Life	1946	
21	22	Seven Samurai	1954	
22	23	The Silence of the Lambs	1991	
23	24	Saving Private Ryan	1998	
24	25	City of God	2002	
25	26	The Green Mile	1999	
26	27	Life Is Beautiful	1997	
27	28	Terminator 2: Judgment Day	1991	
28	29	Star Wars: Episode IV - A New Hope	1977	
29	30	Back to the Future	1985	
30	31	Spirited Away	2001	
31	32	The Pianist	2002	
32	33	Parasite	2019	
33	34	Psycho	1960	
34	35	Gladiator	2000	

35	36	The Lion King	1994
36	37	The Departed	2006
37	38	Spider-Man: Across the Spider-Verse	2023
38	39	Whiplash	2014
39	40	Grave of the Fireflies	1988
40	41	American History X	1998
41	42	Léon: The Professional	1994
42	43	The Prestige	2006
43	44	Harakiri	1962
44	45	Casablanca	1942
45	46	The Usual Suspects	1995
46	47	The Intouchables	2011
47	48	Dune: Part Two	2024
48	49	Cinema Paradiso	1988
49	50	Modern Times	1936
50	51	Alien	1979
51	52	Rear Window	1954
52	53	Once Upon a Time in the West	1968
53	54	Django Unchained	2012
54	55	City Lights	1931
55	56	Apocalypse Now	1979
56	57	Memento	2000
57	58	WALL·E	2008
58	59	Raiders of the Lost Ark	1981
59	60	The Lives of Others	2006

	content_rating	rating	runtime_minutes	rating_scaled	\
0	R	9.3	142	93.0	
1	R	9.2	175	92.0	
2	PG-13	9.0	152	90.0	
3	R	9.0	202	90.0	
4	Approved	9.0	96	90.0	
5	PG-13	9.0	201	90.0	
6	R	9.0	195	90.0	
7	R	8.9	154	89.0	
8	PG-13	8.9	178	89.0	
9	R	8.8	178	88.0	
10	PG-13	8.8	142	88.0	
11	PG-13	8.8	179	88.0	
12	R	8.8	139	88.0	
13	PG-13	8.8	148	88.0	
14	PG	8.7	124	87.0	
15	R	8.7	136	87.0	
16	R	8.7	145	87.0	
17	R	8.7	133	87.0	
18	PG-13	8.7	169	87.0	
19	R	8.6	127	86.0	

20	PG	8.6	130	86.0
21	Not Rated	8.6	207	86.0
22	R	8.6	118	86.0
23	R	8.6	169	86.0
24	R	8.6	130	86.0
25	R	8.6	189	86.0
26	PG-13	8.6	116	86.0
27	R	8.6	137	86.0
28	PG	8.6	121	86.0
29	PG	8.5	116	85.0
30	PG	8.6	124	86.0
31	R	8.5	150	85.0
32	R	8.5	132	85.0
33	R	8.5	109	85.0
34	R	8.5	155	85.0
35	G	8.5	88	85.0
36	R	8.5	151	85.0
37	PG	8.6	140	86.0
38	R	8.5	106	85.0
39	Not Rated	8.5	88	85.0
40	R	8.5	119	85.0
41	R	8.5	110	85.0
42	PG-13	8.5	130	85.0
43	Not Rated	8.6	133	86.0
44	PG	8.5	102	85.0
45	R	8.5	106	85.0
46	R	8.5	112	85.0
47	PG-13	8.5	166	85.0
48	PG	8.5	174	85.0
49	G	8.5	87	85.0
50	R	8.5	117	85.0
51	PG	8.5	112	85.0
52	PG-13	8.5	166	85.0
53	R	8.5	165	85.0
54	G	8.5	87	85.0
55	R	8.4	147	84.0
56	R	8.4	113	84.0
57	G	8.4	98	84.0
58	PG	8.4	115	84.0
59	R	8.4	137	84.0

	title	year	rating	...	\
0	The Shawshank Redemption	1994.0	R	...	
1	The Godfather	1972.0	R	...	
2	The Dark Knight	2008.0	PG-13	...	
3	The Godfather Part II	1974.0	R	...	
4	12 Angry Men	1957.0	Approved	...	

5	The Lord of the Rings: The Return of the King	2003.0	PG-13	...
6	Schindler's List	1994.0	R	...
7	Pulp Fiction	1994.0	R	...
8	The Lord of the Rings: The Fellowship of the Ring	2001.0	PG-13	...
9	The Good, the Bad and the Ugly	1967.0	R	...
10	Forrest Gump	1994.0	PG-13	...
11	The Lord of the Rings: The Two Towers	2002.0	PG-13	...
12	Fight Club	1999.0	R	...
13	Inception	2010.0	PG-13	...
14	Star Wars: Episode V - The Empire Strikes Back	1980.0	PG	...
15	The Matrix	1999.0	R	...
16	None	NaN	None	...
17	One Flew Over the Cuckoo's Nest	1975.0	R	...
18	Interstellar	2014.0	PG-13	...
19	None	NaN	None	...
20	It's a Wonderful Life	1946.0	PG	...
21	Seven Samurai	1954.0	Not Rated	...
22	The Silence of the Lambs	1991.0	R	...
23	Saving Private Ryan	1998.0	R	...
24	City of God	2004.0	R	...
25	The Green Mile	1999.0	R	...
26	Life Is Beautiful	1997.0	PG-13	...
27	Terminator 2: Judgment Day	1991.0	R	...
28	Star Wars: Episode IV - A New Hope	1977.0	PG	...
29	Back to the Future	1985.0	PG	...
30	Spirited Away	2003.0	PG	...
31	The Pianist	2003.0	R	...
32	Parasite	2019.0	R	...
33	None	NaN	None	...
34	Gladiator	2000.0	R	...
35	The Lion King	1994.0	G	...
36	The Departed	2006.0	R	...
37	Spider-Man: Across the Spider-Verse	2023.0	PG	...
38	Whiplash	2014.0	R	...
39	Grave of the Fireflies	1989.0	Not Rated	...
40	American History X	1998.0	R	...
41	Léon: The Professional	1994.0	R	...
42	The Prestige	2006.0	PG-13	...
43	Harakiri	1964.0	Not Rated	...
44	Casablanca	1943.0	PG	...
45	The Usual Suspects	1995.0	R	...
46	The Intouchables	2011.0	R	...
47	Dune: Part Two	2024.0	PG-13	...
48	Cinema Paradiso	1990.0	PG	...
49	Modern Times	1936.0	G	...
50	Alien	1979.0	R	...
51	Rear Window	1954.0	PG	...

52	Once Upon a Time in the West	1969.0	PG-13	...
53	Django Unchained	2012.0	R	...
54	City Lights	1931.0	G	...
55	Apocalypse Now	1979.0	R	...
56	Memento	2001.0	R	...
57	WALL·E	2008.0	G	...
58	Raiders of the Lost Ark	1981.0	PG	...
59	The Lives of Others	2006.0	R	...

	Audience Status	Audience Rating	Audience Count	\
0	Upright	98.0	886485.0	
1	Upright	98.0	733939.0	
2	Upright	94.0	1831566.0	
3	None	NaN	NaN	
4	Upright	87.0	20468.0	
5	Upright	86.0	34679773.0	
6	Upright	97.0	411608.0	
7	Upright	96.0	1127813.0	
8	Upright	95.0	1355751.0	
9	Upright	97.0	239803.0	
10	Upright	95.0	1244237.0	
11	Upright	95.0	1341428.0	
12	Upright	96.0	1093242.0	
13	Upright	91.0	571078.0	
14	Upright	97.0	1058000.0	
15	Upright	85.0	33324202.0	
16	None	NaN	NaN	
17	Upright	96.0	279760.0	
18	Upright	86.0	175957.0	
19	None	NaN	NaN	
20	Upright	95.0	218883.0	
21	None	NaN	NaN	
22	Upright	95.0	847028.0	
23	Upright	95.0	993261.0	
24	None	NaN	NaN	
25	Upright	94.0	813775.0	
26	None	NaN	NaN	
27	Upright	94.0	748984.0	
28	Upright	96.0	862692.0	
29	Upright	94.0	1095020.0	
30	Upright	96.0	336970.0	
31	Upright	96.0	253230.0	
32	Spilled	14.0	2075.0	
33	Spilled	28.0	68326.0	
34	Upright	68.0	9794.0	
35	Upright	93.0	1269333.0	
36	Upright	94.0	737806.0	

37	None	NaN	NaN
38	Upright	94.0	70212.0
39	None	NaN	NaN
40	Upright	96.0	479824.0
41	Upright	95.0	274298.0
42	Upright	92.0	551259.0
43	Upright	97.0	5960.0
44	Upright	95.0	357518.0
45	Upright	96.0	434949.0
46	Upright	93.0	45398.0
47	None	NaN	NaN
48	None	NaN	NaN
49	Upright	95.0	40236.0
50	Upright	94.0	459868.0
51	Upright	95.0	150824.0
52	Upright	95.0	65809.0
53	Upright	91.0	429231.0
54	Upright	96.0	27080.0
55	Upright	94.0	286037.0
56	Upright	94.0	381037.0
57	None	NaN	NaN
58	Upright	96.0	826663.0
59	Upright	96.0	121677.0

	Tomatometer Top Critics Count	Tomatometer Fresh Critics Count \
0	19.0	68.0
1	24.0	97.0
2	63.0	320.0
3	NaN	NaN
4	0.0	11.0
5	53.0	257.0
6	24.0	97.0
7	32.0	97.0
8	48.0	213.0
9	15.0	72.0
10	30.0	73.0
11	52.0	243.0
12	40.0	137.0
13	65.0	310.0
14	29.0	97.0
15	37.0	135.0
16	NaN	NaN
17	14.0	75.0
18	62.0	263.0
19	NaN	NaN
20	16.0	77.0
21	NaN	NaN

22	28.0	97.0
23	42.0	132.0
24	NaN	NaN
25	35.0	105.0
26	NaN	NaN
27	26.0	76.0
28	45.0	120.0
29	23.0	81.0
30	45.0	182.0
31	40.0	173.0
32	3.0	2.0
33	18.0	30.0
34	6.0	4.0
35	29.0	120.0
36	60.0	254.0
37	NaN	NaN
38	67.0	277.0
39	NaN	NaN
40	23.0	71.0
41	18.0	48.0
42	48.0	152.0
43	2.0	8.0
44	23.0	91.0
45	22.0	68.0
46	40.0	94.0
47	NaN	NaN
48	NaN	NaN
49	13.0	61.0
50	29.0	122.0
51	19.0	73.0
52	15.0	61.0
53	55.0	250.0
54	13.0	51.0
55	25.0	93.0
56	43.0	166.0
57	NaN	NaN
58	19.0	76.0
59	54.0	149.0

	Tomatometer	Rotten Critics Count	z_score_Tomatometer	Rating \
0		7.0		1.055008
1		2.0		1.301327
2		20.0		1.160573
3		NaN		NaN
4		1.0		1.090197
5		18.0		1.125385
6		3.0		1.266138

7	9.0	1.090197
8	20.0	1.055008
9	2.0	1.266138
10	30.0	0.351242
11	12.0	1.195762
12	36.0	0.632748
13	47.0	0.914255
14	6.0	1.160573
15	19.0	0.949443
16	NaN	NaN
17	5.0	1.160573
18	101.0	0.386430
19	NaN	NaN
20	5.0	1.160573
21	NaN	NaN
22	4.0	1.230950
23	10.0	1.125385
24	NaN	NaN
25	29.0	0.597560
26	NaN	NaN
27	6.0	1.125385
28	10.0	1.090197
29	3.0	1.230950
30	5.0	1.266138
31	9.0	1.195762
32	9.0	-1.513740
33	48.0	-0.809974
34	9.0	-1.056292
35	9.0	1.125385
36	26.0	1.055008
37	NaN	NaN
38	19.0	1.160573
39	NaN	NaN
40	15.0	0.773502
41	17.0	0.456807
42	49.0	0.527183
43	0.0	1.371703
44	1.0	1.336515
45	8.0	0.984632
46	32.0	0.491995
47	NaN	NaN
48	NaN	NaN
49	0.0	1.371703
50	3.0	1.301327
51	1.0	1.336515
52	3.0	1.195762
53	38.0	0.914255

54	1.0	1.301327
55	2.0	1.301327
56	13.0	1.125385
57	NaN	NaN
58	4.0	1.195762
59	12.0	1.125385

	z_score_Audience Rating	z_score_Tomatometer Count	z_score_Audience Count
0	1.830037	0.282118	0.452609
1	1.830037	0.638966	0.360994
2	1.633725	4.222318	1.020201
3	NaN	NaN	NaN
4	1.290180	-0.654609	-0.067500
5	1.241102	3.255854	20.748027
6	1.780959	0.653835	0.167410
7	1.731881	0.743047	0.597545
8	1.682803	2.631369	0.734439
9	1.780959	0.267249	0.064228
10	1.682803	0.698441	0.667466
11	1.682803	2.958480	0.725837
12	1.731881	1.739249	0.576782
13	1.486492	4.475085	0.263183
14	1.780959	0.698441	0.555617
15	1.192024	1.456744	19.933905
16	NaN	NaN	NaN
17	1.731881	0.356461	0.088225
18	1.241102	4.579166	0.025883
19	NaN	NaN	NaN
20	1.682803	0.386199	0.051664
21	NaN	NaN	NaN
22	1.682803	0.668704	0.428912
23	1.682803	1.278319	0.516736
24	NaN	NaN	NaN
25	1.633725	1.159370	0.408941
26	NaN	NaN	NaN
27	1.633725	0.386199	0.370029
28	1.731881	1.099895	0.438319
29	1.633725	0.415936	0.577850
30	1.731881	1.947410	0.122584
31	1.731881	1.873067	0.072292
32	-2.292503	-0.669477	-0.078546
33	-1.605413	0.326724	-0.038757
34	0.357701	-0.639740	-0.073910
35	1.584647	1.085027	0.682538
36	1.633725	3.330197	0.363316
37	NaN	NaN	NaN
38	1.633725	3.568096	-0.037625

39	NaN	NaN	NaN
40	1.731881	0.445674	0.208378
41	1.682803	0.133431	0.084945
42	1.535569	2.155571	0.251280
43	1.780959	-0.714083	-0.076213
44	1.682803	0.534886	0.134924
45	1.731881	0.296987	0.181428
46	1.584647	1.040421	-0.052527
47	NaN	NaN	NaN
48	NaN	NaN	NaN
49	1.682803	0.073957	-0.055627
50	1.633725	1.025552	0.196393
51	1.682803	0.267249	0.010789
52	1.682803	0.118563	-0.040269
53	1.486492	3.449146	0.177993
54	1.731881	-0.059862	-0.063529
55	1.633725	0.579492	0.091995
56	1.633725	1.828461	0.149049
57	NaN	NaN	NaN
58	1.731881	0.356461	0.416681
59	1.731881	1.560824	-0.006716

[60 rows x 51 columns]

```
[14]: joined_movies_dataset.head(60)
```

```
[14]:
```

	rank	title	release_year	\
0	1	The Shawshank Redemption	1994	
1	2	The Godfather	1972	
2	3	The Dark Knight	2008	
3	4	The Godfather Part II	1974	
4	5	12 Angry Men	1957	
5	6	The Lord of the Rings: The Return of the King	2003	
6	7	Schindler's List	1993	
7	8	Pulp Fiction	1994	
8	9	The Lord of the Rings: The Fellowship of the Ring	2001	
9	10	The Good, the Bad and the Ugly	1966	
10	11	Forrest Gump	1994	
11	12	The Lord of the Rings: The Two Towers	2002	
12	13	Fight Club	1999	
13	14	Inception	2010	
14	15	Star Wars: Episode V - The Empire Strikes Back	1980	
15	16	The Matrix	1999	
16	17	Goodfellas	1990	
17	18	One Flew Over the Cuckoo's Nest	1975	
18	19	Interstellar	2014	
19	20	Se7en	1995	

20	21	It's a Wonderful Life	1946
21	22	Seven Samurai	1954
22	23	The Silence of the Lambs	1991
23	24	Saving Private Ryan	1998
24	25	City of God	2002
25	26	The Green Mile	1999
26	27	Life Is Beautiful	1997
27	28	Terminator 2: Judgment Day	1991
28	29	Star Wars: Episode IV - A New Hope	1977
29	30	Back to the Future	1985
30	31	Spirited Away	2001
31	32	The Pianist	2002
32	33	Parasite	2019
33	34	Psycho	1960
34	35	Gladiator	2000
35	36	The Lion King	1994
36	37	The Departed	2006
37	38	Spider-Man: Across the Spider-Verse	2023
38	39	Whiplash	2014
39	40	Grave of the Fireflies	1988
40	41	American History X	1998
41	42	Léon: The Professional	1994
42	43	The Prestige	2006
43	44	Harakiri	1962
44	45	Casablanca	1942
45	46	The Usual Suspects	1995
46	47	The Intouchables	2011
47	48	Dune: Part Two	2024
48	49	Cinema Paradiso	1988
49	50	Modern Times	1936
50	51	Alien	1979
51	52	Rear Window	1954
52	53	Once Upon a Time in the West	1968
53	54	Django Unchained	2012
54	55	City Lights	1931
55	56	Apocalypse Now	1979
56	57	Memento	2000
57	58	WALL·E	2008
58	59	Raiders of the Lost Ark	1981
59	60	The Lives of Others	2006

	content_rating	rating	runtime_minutes	rating_scaled \
0	R	9.3	142	93.0
1	R	9.2	175	92.0
2	PG-13	9.0	152	90.0
3	R	9.0	202	90.0
4	Approved	9.0	96	90.0

5	PG-13	9.0	201	90.0
6	R	9.0	195	90.0
7	R	8.9	154	89.0
8	PG-13	8.9	178	89.0
9	R	8.8	178	88.0
10	PG-13	8.8	142	88.0
11	PG-13	8.8	179	88.0
12	R	8.8	139	88.0
13	PG-13	8.8	148	88.0
14	PG	8.7	124	87.0
15	R	8.7	136	87.0
16	R	8.7	145	87.0
17	R	8.7	133	87.0
18	PG-13	8.7	169	87.0
19	R	8.6	127	86.0
20	PG	8.6	130	86.0
21	Not Rated	8.6	207	86.0
22	R	8.6	118	86.0
23	R	8.6	169	86.0
24	R	8.6	130	86.0
25	R	8.6	189	86.0
26	PG-13	8.6	116	86.0
27	R	8.6	137	86.0
28	PG	8.6	121	86.0
29	PG	8.5	116	85.0
30	PG	8.6	124	86.0
31	R	8.5	150	85.0
32	R	8.5	132	85.0
33	R	8.5	109	85.0
34	R	8.5	155	85.0
35	G	8.5	88	85.0
36	R	8.5	151	85.0
37	PG	8.6	140	86.0
38	R	8.5	106	85.0
39	Not Rated	8.5	88	85.0
40	R	8.5	119	85.0
41	R	8.5	110	85.0
42	PG-13	8.5	130	85.0
43	Not Rated	8.6	133	86.0
44	PG	8.5	102	85.0
45	R	8.5	106	85.0
46	R	8.5	112	85.0
47	PG-13	8.5	166	85.0
48	PG	8.5	174	85.0
49	G	8.5	87	85.0
50	R	8.5	117	85.0
51	PG	8.5	112	85.0

52	PG-13	8.5	166	85.0
53	R	8.5	165	85.0
54	G	8.5	87	85.0
55	R	8.4	147	84.0
56	R	8.4	113	84.0
57	G	8.4	98	84.0
58	PG	8.4	115	84.0
59	R	8.4	137	84.0

		title.1	year	rating.1	...	\
0		The Shawshank Redemption	1994.0	R	...	
1		The Godfather	1972.0	R	...	
2		The Dark Knight	2008.0	PG-13	...	
3		The Godfather Part II	1974.0	R	...	
4		12 Angry Men	1957.0	Approved	...	
5	The Lord of the Rings: The Return of the King		2003.0	PG-13	...	
6		Schindler's List	1994.0	R	...	
7		Pulp Fiction	1994.0	R	...	
8	The Lord of the Rings: The Fellowship of the Ring		2001.0	PG-13	...	
9		The Good, the Bad and the Ugly	1967.0	R	...	
10		Forrest Gump	1994.0	PG-13	...	
11		The Lord of the Rings: The Two Towers	2002.0	PG-13	...	
12		Fight Club	1999.0	R	...	
13		Inception	2010.0	PG-13	...	
14	Star Wars: Episode V - The Empire Strikes Back		1980.0	PG	...	
15		The Matrix	1999.0	R	...	
16		NaN	NaN	NaN	...	
17		One Flew Over the Cuckoo's Nest	1975.0	R	...	
18		Interstellar	2014.0	PG-13	...	
19		NaN	NaN	NaN	...	
20		It's a Wonderful Life	1946.0	PG	...	
21		Seven Samurai	1954.0	Not Rated	...	
22		The Silence of the Lambs	1991.0	R	...	
23		Saving Private Ryan	1998.0	R	...	
24		City of God	2004.0	R	...	
25		The Green Mile	1999.0	R	...	
26		Life Is Beautiful	1997.0	PG-13	...	
27		Terminator 2: Judgment Day	1991.0	R	...	
28	Star Wars: Episode IV - A New Hope		1977.0	PG	...	
29		Back to the Future	1985.0	PG	...	
30		Spirited Away	2003.0	PG	...	
31		The Pianist	2003.0	R	...	
32		Parasite	2019.0	R	...	
33		NaN	NaN	NaN	...	
34		Gladiator	2000.0	R	...	
35		The Lion King	1994.0	G	...	
36		The Departed	2006.0	R	...	

37	Spider-Man: Across the Spider-Verse	2023.0	PG	...
38	Whiplash	2014.0	R	...
39	Grave of the Fireflies	1989.0	Not Rated	...
40	American History X	1998.0	R	...
41	Léon: The Professional	1994.0	R	...
42	The Prestige	2006.0	PG-13	...
43	Harakiri	1964.0	Not Rated	...
44	Casablanca	1943.0	PG	...
45	The Usual Suspects	1995.0	R	...
46	The Intouchables	2011.0	R	...
47	Dune: Part Two	2024.0	PG-13	...
48	Cinema Paradiso	1990.0	PG	...
49	Modern Times	1936.0	G	...
50	Alien	1979.0	R	...
51	Rear Window	1954.0	PG	...
52	Once Upon a Time in the West	1969.0	PG-13	...
53	Django Unchained	2012.0	R	...
54	City Lights	1931.0	G	...
55	Apocalypse Now	1979.0	R	...
56	Memento	2001.0	R	...
57	WALL·E	2008.0	G	...
58	Raiders of the Lost Ark	1981.0	PG	...
59	The Lives of Others	2006.0	R	...

	Audience Rating	Audience Count	Tomatometer	Top Critics Count	\
0	98.0	886485.0		19.0	
1	98.0	733939.0		24.0	
2	94.0	1831566.0		63.0	
3	NaN	NaN		NaN	
4	87.0	20468.0		0.0	
5	86.0	34679773.0		53.0	
6	97.0	411608.0		24.0	
7	96.0	1127813.0		32.0	
8	95.0	1355751.0		48.0	
9	97.0	239803.0		15.0	
10	95.0	1244237.0		30.0	
11	95.0	1341428.0		52.0	
12	96.0	1093242.0		40.0	
13	91.0	571078.0		65.0	
14	97.0	1058000.0		29.0	
15	85.0	33324202.0		37.0	
16	NaN	NaN		NaN	
17	96.0	279760.0		14.0	
18	86.0	175957.0		62.0	
19	NaN	NaN		NaN	
20	95.0	218883.0		16.0	
21	NaN	NaN		NaN	

22	95.0	847028.0	28.0
23	95.0	993261.0	42.0
24	NaN	NaN	NaN
25	94.0	813775.0	35.0
26	NaN	NaN	NaN
27	94.0	748984.0	26.0
28	96.0	862692.0	45.0
29	94.0	1095020.0	23.0
30	96.0	336970.0	45.0
31	96.0	253230.0	40.0
32	14.0	2075.0	3.0
33	28.0	68326.0	18.0
34	68.0	9794.0	6.0
35	93.0	1269333.0	29.0
36	94.0	737806.0	60.0
37	NaN	NaN	NaN
38	94.0	70212.0	67.0
39	NaN	NaN	NaN
40	96.0	479824.0	23.0
41	95.0	274298.0	18.0
42	92.0	551259.0	48.0
43	97.0	5960.0	2.0
44	95.0	357518.0	23.0
45	96.0	434949.0	22.0
46	93.0	45398.0	40.0
47	NaN	NaN	NaN
48	NaN	NaN	NaN
49	95.0	40236.0	13.0
50	94.0	459868.0	29.0
51	95.0	150824.0	19.0
52	95.0	65809.0	15.0
53	91.0	429231.0	55.0
54	96.0	27080.0	13.0
55	94.0	286037.0	25.0
56	94.0	381037.0	43.0
57	NaN	NaN	NaN
58	96.0	826663.0	19.0
59	96.0	121677.0	54.0

	Tomatometer Fresh Critics Count	Tomatometer Rotten Critics Count \
0	68.0	7.0
1	97.0	2.0
2	320.0	20.0
3	NaN	NaN
4	11.0	1.0
5	257.0	18.0
6	97.0	3.0

7	97.0	9.0
8	213.0	20.0
9	72.0	2.0
10	73.0	30.0
11	243.0	12.0
12	137.0	36.0
13	310.0	47.0
14	97.0	6.0
15	135.0	19.0
16	NaN	NaN
17	75.0	5.0
18	263.0	101.0
19	NaN	NaN
20	77.0	5.0
21	NaN	NaN
22	97.0	4.0
23	132.0	10.0
24	NaN	NaN
25	105.0	29.0
26	NaN	NaN
27	76.0	6.0
28	120.0	10.0
29	81.0	3.0
30	182.0	5.0
31	173.0	9.0
32	2.0	9.0
33	30.0	48.0
34	4.0	9.0
35	120.0	9.0
36	254.0	26.0
37	NaN	NaN
38	277.0	19.0
39	NaN	NaN
40	71.0	15.0
41	48.0	17.0
42	152.0	49.0
43	8.0	0.0
44	91.0	1.0
45	68.0	8.0
46	94.0	32.0
47	NaN	NaN
48	NaN	NaN
49	61.0	0.0
50	122.0	3.0
51	73.0	1.0
52	61.0	3.0
53	250.0	38.0

54	51.0	1.0
55	93.0	2.0
56	166.0	13.0
57	NaN	NaN
58	76.0	4.0
59	149.0	12.0

	z_score_TomatoMeter Rating	z_score_Audience Rating \
0	1.055008	1.830037
1	1.301327	1.830037
2	1.160573	1.633725
3	NaN	NaN
4	1.090197	1.290180
5	1.125385	1.241102
6	1.266138	1.780959
7	1.090197	1.731881
8	1.055008	1.682803
9	1.266138	1.780959
10	0.351242	1.682803
11	1.195762	1.682803
12	0.632748	1.731881
13	0.914255	1.486492
14	1.160573	1.780959
15	0.949443	1.192024
16	NaN	NaN
17	1.160573	1.731881
18	0.386430	1.241102
19	NaN	NaN
20	1.160573	1.682803
21	NaN	NaN
22	1.230950	1.682803
23	1.125385	1.682803
24	NaN	NaN
25	0.597560	1.633725
26	NaN	NaN
27	1.125385	1.633725
28	1.090197	1.731881
29	1.230950	1.633725
30	1.266138	1.731881
31	1.195762	1.731881
32	-1.513740	-2.292503
33	-0.809974	-1.605413
34	-1.056292	0.357701
35	1.125385	1.584647
36	1.055008	1.633725
37	NaN	NaN
38	1.160573	1.633725

39	NaN	NaN
40	0.773502	1.731881
41	0.456807	1.682803
42	0.527183	1.535569
43	1.371703	1.780959
44	1.336515	1.682803
45	0.984632	1.731881
46	0.491995	1.584647
47	NaN	NaN
48	NaN	NaN
49	1.371703	1.682803
50	1.301327	1.633725
51	1.336515	1.682803
52	1.195762	1.682803
53	0.914255	1.486492
54	1.301327	1.731881
55	1.301327	1.633725
56	1.125385	1.633725
57	NaN	NaN
58	1.195762	1.731881
59	1.125385	1.731881

	z_score_TomatoMeter Count	z_score_Audience Count	title_sentiment
0	0.282118	0.452609	0.000000
1	0.638966	0.360994	0.000000
2	4.222318	1.020201	-0.150000
3	NaN	NaN	0.000000
4	-0.654609	-0.067500	-0.500000
5	3.255854	20.748027	0.000000
6	0.653835	0.167410	0.000000
7	0.743047	0.597545	0.000000
8	2.631369	0.734439	0.000000
9	0.267249	0.064228	-0.233333
10	0.698441	0.667466	0.000000
11	2.958480	0.725837	0.000000
12	1.739249	0.576782	0.000000
13	4.475085	0.263183	0.000000
14	0.698441	0.555617	0.000000
15	1.456744	19.933905	0.000000
16	NaN	NaN	0.000000
17	0.356461	0.088225	0.000000
18	4.579166	0.025883	0.000000
19	NaN	NaN	0.000000
20	0.386199	0.051664	1.000000
21	NaN	NaN	0.000000
22	0.668704	0.428912	0.000000
23	1.278319	0.516736	0.000000

24	NaN	NaN	0.000000
25	1.159370	0.408941	-0.200000
26	NaN	NaN	0.850000
27	0.386199	0.370029	0.000000
28	1.099895	0.438319	0.136364
29	0.415936	0.577850	0.000000
30	1.947410	0.122584	0.500000
31	1.873067	0.072292	0.000000
32	-0.669477	-0.078546	0.000000
33	0.326724	-0.038757	0.000000
34	-0.639740	-0.073910	0.000000
35	1.085027	0.682538	0.000000
36	3.330197	0.363316	0.000000
37	NaN	NaN	0.000000
38	3.568096	-0.037625	0.000000
39	NaN	NaN	0.000000
40	0.445674	0.208378	0.000000
41	0.133431	0.084945	0.100000
42	2.155571	0.251280	0.000000
43	-0.714083	-0.076213	0.000000
44	0.534886	0.134924	0.000000
45	0.296987	0.181428	-0.250000
46	1.040421	-0.052527	0.000000
47	NaN	NaN	0.000000
48	NaN	NaN	0.000000
49	0.073957	-0.055627	0.200000
50	1.025552	0.196393	-0.250000
51	0.267249	0.010789	0.000000
52	0.118563	-0.040269	0.000000
53	3.449146	0.177993	0.000000
54	-0.059862	-0.063529	0.000000
55	0.579492	0.091995	0.000000
56	1.828461	0.149049	0.000000
57	NaN	NaN	0.000000
58	0.356461	0.416681	0.000000
59	1.560824	-0.006716	0.000000

[60 rows x 52 columns]